

Observing the Memory Mills Running

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Kernel Recipes 2025

Last year...

We looked at /proc/meminfo

Legend: Kernel Userspace *Redundant* Not consumed memory **Sums to MemTotal**

MemTotal:	32125696 kB	SwapFree:	2513056 kB	CommitLimit:	18575904 kB
MemFree:	**8627220 kB**	**Zswap:**	**0 kB**	Committed_AS:	21267524 kB
MemAvailable:	21107232 kB	Zswapped:	0 kB	**VmallocUsed:**	**84632 kB**
Buffers:	**258820 kB**	*Dirty:*	*844 kB*	**Percpu:**	**7680 kB**
Cached:	**12902952 kB**	*Writeback:*	*196 kB*	*AnonHugePages:*	*1363968 kB*
SwapCached:	*0 kB*	**AnonPages:**	**8922944 kB**	*ShmemHugePages:*	*0 kB*
Active:	*5870672 kB*	*Mapped:*	*1683592 kB*	*ShmemPmdMapped:*	*0 kB*
Inactive:	*16277940 kB*	*Shmem:*	*611016 kB*	*FileHugePages:*	*1062912 kB*
Active(anon):	*8120 kB*	*KReclaimable:*	*390056 kB*	*FilePmdMapped:*	*684032 kB*
Inactive(anon):	*9589756 kB*	**Slab:**	**613748 kB**	CmaTotal:	0 kB
Active(file):	*5862552 kB*	*SReclaimable:*	*390056 kB*	CmaFree:	0 kB
Inactive(file):	*6688184 kB*	*SUnreclaim:*	*223692 kB*	Balloon:	0 kB
Unevictable:	*80 kB*	**KernelStack:**	**35568 kB**	**Hugetlb:**	**0 kB**
Mlocked:	*80 kB*	**PageTables:**	**82500 kB**		
SwapTotal:	2513056 kB	**SecPageTables:**	**3248 kB**		

Motivation: what to make of memory usage

- "`MemFree`" isn't a useful value, will converge to near-zero eventually on most systems
- "`MemAvailable`" and "*used*" (reported by `free (1)` as *total* minus *available*) make much more sense to report the system state at any given moment
- during the workload's steady operation, the exact values are rarely important
- values may change on distro upgrades due to kernel/glibc implementation details
  - kernel/user memory usage – changes e.g. due to scalability, sizing of structures
  - active/inactive LRU sizes – reclaim algorithm may change, doesn't say much about the workload
- limits for memcgs / alerts determined on an older version may need updating
  - memcg kmem consumption – new kinds of kernel objects may get accounted

Motivation: what to make of memory usage

- What matters more than exact values - trends over time or dynamic changes
  - a particular counter keeps growing over time - possible memory leak?
  - “available” memory exhausted, only a little page cache remaining, thrashing or OOM kills?
  - is this due to to kernel bug, userspace bug or my workload is simply too big?
- Looking at `/proc/vmstat` event counters can tell much more information and often explain the changes in memory counters
  - See what actions the kernel is doing to keep up with the memory demand
  - But what exactly are the events that we can observe? Topic for today!
    - vmstat has 193 lines (meminfo only 58) – hope you have enough time…
      - Just kidding, let’s leave that to a 1300 pages long book ;)

/proc/vmstat at a glance

- /proc/meminfo is curated and formatted for humans
  - kB, names, ordering, values calculated from multiple counters
- /proc/vmstat is more suitable for script consumption
  - Mostly a dump of simple counter names in their "enum" order and their raw values
  - Some counters ("**nr**_") are for state, can increase and decrease, many used also in /proc/meminfo
  - Others (not "**nr**_") count events (since boot) and thus can only increment
  - But we are talking about the Linux kernel (and memory management) here, so of course the naming is not 100% consistent and there are exceptions in both directions
  - Some are internally tracked per NUMA node or even zone (DMA, DMA32, Normal…) and summed up for /proc/vmstat
    - Node-specific subsets in /sys/bus/node/devices/nodeX/vmstat
    - Per-zone breakdown in /proc/zoneinfo

```
nr_free_pages 241422
nr_free_pages_blocks 113152
nr_zone_inactive_anon 2566933
nr_zone_active_anon 106115
nr_zone_inactive_file 775357
nr_zone_active_file 3716294
nr_zone_unevictable 120
nr_zone_write_pending 2454
nr_mlock 120
nr_zspages 0
nr_free_cma 0
nr_unaccepted 0
numa_hit 119608711
numa_miss 0
numa_foreign 0
numa_interleave 2979
numa_local 119608711
numa_other 0
nr_inactive_anon 2566933
nr_active_anon 106115
nr_inactive_file 775357
nr_active_file 3716294
nr_unevictable 120
nr_slab_reclaimable 283532
nr_slab_unreclaimable 86648
nr_isolated_anon 0
nr_isolated_file 0
workingset_nodes 82120
workingset_refault_anon 80
workingset_refault_file 741707
workingset_activate_anon 16
workingset_activate_file 372224
workingset_restore_anon 0
workingset_restore_file 15228
workingset_nodereclaim 0
nr_anon_pages 2650475
nr_mapped 480776
nr_file_pages 4478585
nr_dirty 2454
nr_writeback 0
nr_shmem 214152
nr_shmem_hugepages 0
nr_shmem_pmdmapped 0
nr_file_hugepages 863
nr_file_pmdmapped 270
nr_anon_transparent_hugepages 638
nr_vmscan_write 2236
nr_vmscan_immediate_reclaim 42309
nr_dirtied 9182682
nr_written 8931323
nr_throttled_written 0
nr_kernel_misc_reclaimable 0
nr_foll_pin_acquired 3108
nr_foll_pin_released 3108
nr_kernel_stack 44176
nr_page_table_pages 25962
nr_sec_page_table_pages 581
nr_iommu_pages 581
nr_swapcached 31
pgpromote_success 0
pgpromote_candidate 0
pgdemote_kswapd 0
pgdemote_direct 0
pgdemote_khugepaged 0
pgdemote_proactive 0
nr_hugetlb 0
```


vmstat 1/3

```
nr_free_pages 241422
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nr_sl
n
nr_
nr_is
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pgdemote_proactive 0
nr_hugetlb 0
```

enum zone_stat_item
per-zone state

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nr_free_pages_blocks 113152
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nr_vmscan_write 2236
nr_vmscan_immediate_reclaim 42309
nr_dirtied 9182682
nr_written 8931323
```

enum numa_stat_item
per-zone events
related to page allocation
(not access)
per-node:
/sys/.../nodeX/numastat
can be dislabled:
vm.numa_stat sysctl

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nr_zone_active_anon 106115
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nr_dirty 2454
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pgdemote_khugepaged 0
pgdemote_proactive 0
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nr_balloon_pages 0
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enum node_stat_item
per-node
both state and events

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nr_inactive_anon 2566933
nr_active_anon 106115
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pgdemote_proactive 0
nr_hugetlb 0
nr_balloon_pages 0
```

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nr_zone_inactive_file 775357
nr_zone_active_file 3716294
nr_zone_unevictable 120
nr_zone_write_pending 2454
nr_mlock 120
nr_zspages 0
nr_free_cma 0
nr_unaccepted 0
```

enum node_stat_item
per-node
both state and events

```
nr_inactive_anon 2566933
nr_active_anon 106115
nr_inactive_file 775357
nr_active_file 3716294
```

duplicates the
per-zone state

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nr_unevictable 120
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nr_iommu_pages 581
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pgpromote_candidate 0
pgdemote_kswapd 0
pgdemote_direct 0
pgdemote_khugepaged 0
pgdemote_proactive 0
nr_hugetlb 0
nr_balloon_pages 0
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nr_zone_active_file 3716294
nr_zone_unevictable 120
nr_zone_write_pending 2454
nr_mlock 120
nr_zspages 0
nr_free_cma 0
nr_unaccepted 0
```

```
nr_unevictable 120
nr_slab_reclaimable 28353
nr_slab_unreclaimable 866
nr_isolated_anon 0
nr_isolated_file 0
workingset_nodes 82120
workingset_refault_anon 80
workingset_refault_file 741707
                        16
                        372224
                        0
workingset_restore_file 15228
rkingset_nodereclaim 0
_anon_pages 2650475
_mapped 480776
ile_pages 4478585
_dirty 2454
r_writeback 0
```

```
270
nr_anon_transparent_hugepages 638
nr_vmscan_write 2236
nr_vmscan_immediate_reclaim 42309
nr_dirtied 9182682
nr_written 8931323
nr_throttled_written 0
nr_kernel_misc_reclaimable 0
nr_foll_pin_acquired 3108
nr_foll_pin_released 3108
nr_kernel_stack 44176
nr_page_table_pages 25962
nr_sec_page_table_pages 581
nr_iommu_pages 581
nr_swapcached 31
pgpromote_success 0
pgpromote_candidate 0
pgdemote_kswapd 0
pgdemote_direct 0
pgdemote_khugepaged 0
pgdemote_proactive 0
nr_hugetlb 0
nr_balloon_pages 0
```

state without “nr_”

events with “nr_”

enum node_stat_item
per-node
both state and events

```
nr_inactive_anon 2566933
nr_active_anon 106115
nr_inactive_file 775357
nr_active_file 3716294
```

duplicates the
per-zone state

```
nr_balloon_pages 0
nr_dirty_threshold 933288
nr_dirty_background_threshold 477418
nr_memmap_pages 32768
nr_memmap_boot_pages 131072
pgpgin 24575914
pgpgout 36978233
pswpin 5
pswpout 944
pgalloc_dma 1024
pgalloc_dma32 94618536
pgalloc_normal 196507807
pgalloc_movable 0
pgalloc_device 0
allocstall_dma 0
allocstall_dma32 0
allocstall_normal 61
allocstall_movable 318
allocstall_device 0
pgskip_dma 0
pgskip_dma32 0
pgskip_normal 485892
pgskip_movable 0
pgskip_device 0
pgfree 209972695
pgde
pgla
pgfa
pgmajfault 14324
pglazyfreed 538420
pgrefill 256165
pgreuse 1843347
pgsteal_kswapd 4921723
pgsteal_direct 150388
pgsteal_khugepaged 0
pgsteal_proactive 0
pgscan_kswapd 6336189
pgscan_direct 178479
pgscan_khugepaged 0
pgscan_proactive 0
pgscan_direct_throttle 0
pgscan_anon 195515
pgscan_file 6319153
pgsteal_anon 4715
pgsteal_file 5067396
zone_reclaim_success 0
_failed 0
0
d 1428161
steal 21862
kswapd_low_wmark_hit_quickly 1353
kswapd_high_wmark_hit_quickly 47
pageoutrun 1486
pgrotated 52280
drop_pagecache 0
drop_slab 0
oom_kill 0
numa_pte_updates 0
numa_huge_pte_updates 0
numa_hint_faults 0
numa_hint_faults_local 0
numa_pages_migrated 0
pgmigrate_success 1206390
pgmigrate_fail 41740
thp_migration_success 0
```

enum vm_stat_item
global state

enum vm_event_item
global events

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nr_memmap_boot_pages 131072

```
pgpgin 24575914
pgpgout 36978233
pswpin 5
pswpout 944
pgalloc_dma 1024
pgalloc_dma32 94618536
pgalloc_normal 196507807
pgalloc_movable 0
pgalloc_device 0
allocstall_dma 0
allocstall_dma32 0
allocstall_normal 61
allocstall_movable 318
allocstall_device 0
pgskip_dma 0
pgskip_dma32 0
pgskip_normal 485892
```

```
pgskip_movable 0
pgskip_device 0
pgfree 292872685
pgactivate 21200950
pgdeactivate 245777
pglazyfree 1684484
pgfault 105496005
pgmajfault 14324
pglazyfreed 538420
pgrefill 256165
pgreuse 1843347
pgsteal_kswapd 4921723
pgsteal_direct 150388
pgsteal_khugepaged 0
pgsteal_proactive 0
pgscan_kswapd 6336189
pgscan_direct 178479
pgscan_khugepaged 0
pgscan_proactive 0
pgscan_direct_throttle 0
pgscan_anon 195515
pgscan_file 6319153
pgsteal_anon 4715
pgsteal_file 5067396
zone_reclaim_success 0
zone_reclaim_failed 0
pginodesteal 0
slabs_scanned 1428161
kswapd_inodesteal 21862
kswapd_low_wmark_hit_quickly 1353
kswapd_high_wmark_hit_quickly 47
pageoutrun 1486
pgrotated 52280
drop_pagecache 0
drop_slab 0
oom_kill 0
numa_pte_updates 0
numa_huge_pte_updates 0
numa_hint_faults 0
numa_hint_faults_local 0
numa_pages_migrated 0
pgmigrate_success 1206390
pgmigrate_fail 41740
thp_migration_success 0
```

```
thp_migration_fail 0
thp_migration_split 0
compact_migrate_scanned 8863968
compact_free_scanned 86032169
compact_isolated 2509946
compact_stall 1107
compact_fail 636
compact_success 471
compact_daemon_wake 972
compact_daemon_migrate_scanned 8420
compact_daemon_free_scanned 82677
htlb_buddy_alloc_success 0
htlb_buddy_alloc_fail 0
cma_alloc_success 0
cma_alloc_fail 0
unevictable_pgs_culled 131176
unevictable_pgs_scanned 768
unevictable_pgs_rescued 106697
unevictable_pgs_mlocked 106059
unevictable_pgs_munlocked 105883
unevictable_pgs_cleared 0
unevictable_pgs_stranded 56
thp_fault_alloc 265372
thp_fault_fallback 15214
thp_fault_fallback_charge 0
thp_collapse_alloc 959
thp_collapse_alloc_failed 0
thp_file_alloc 0
thp_file_fallback 0
thp_file_fallback_charge 0
thp_file_mapped 32052
thp_split_page 17
thp_split_page_failed 0
thp_deferred_split_page 624
thp_underused_split_page 0
thp_split_pmd 827
thp_scan_exceed_none_pte 1161
thp_scan_exceed_swap_pte 4883
thp_scan_exceed_share_pte 1
thp_split_pud 0
thp_zero_page_alloc 1
thp_zero_page_alloc_failed 0
thp_swpout 0
thp_swpout_fallback 0
balloon_inflate 0
balloon_deflate 0
balloon_migrate 0
swap_ra 67
swap_ra_hit 54
swpin_zero 75
swpout_zero 1292
ksm_swpin_copy 0
cow_ksm 0
zswpin 0
zswpout 0
zswpwb 0
direct_map_level2_splits 411
direct_map_level3_splits 24
direct_map_level2_collapses 0
direct_map_level3_collapses 0
nr_unstable 0
```

vmstat 3/3

enum vm_event_item
global events
(continued)

```
compact_fail 636
compact_success 471
compact_daemon_wake 972
compact_daemon_migrate_scanned 8420
compact_daemon_free_scanned 82677
htlb_buddy_alloc_success 0
htlb_buddy_alloc_fail 0
cma_alloc_success 0
cma_alloc_fail 0
unevictable_pgs_culled 131176
unevictable_pgs_scanned 768
unevictable_pgs_rescued 106697
unevictable_pgs_mlocked 106059
unevictable_pgs_munlocked 105883
unevictable_pgs_cleared 0
unevictable_pgs_stranded 56
thp_fault_alloc 265372
thp_fault_fallback 15214
thp_fault_fallback_charge 0
thp_collapse_alloc 959
thp_collapse_alloc_failed 0
thp_file_alloc 0
thp_file_fallback 0
thp_file_fallback_charge 0
thp_file_mapped 32052
thp_split_page 17
thp_split_page_failed 0
thp_deferred_split_page 624
thp_underused_split_page 0
thp_split_pmd 827
thp_scan_exceed_none_pte 1161
thp_scan_exceed_swap_pte 4883
thp_scan_exceed_share_pte 1
thp_split_pud 0
thp_zero_page_alloc 1
thp_zero_page_alloc_failed 0
thp_swpout 0
thp_swpout_fallback 0
balloon_inflate 0
balloon_deflate 0
balloon_migrate 0
swap_ra 67
swap_ra_hit 54
swpin_zero 75
swpout_zero 1292
ksm_swpin_copy 0
cow_ksm 0
zswpin 0
zswpout 0
zswpwb 0
direct_map_level2_splits 411
direct_map_level3_splits 24
direct_map_level2_collapses 0
direct_map_level3_collapses 0
nr_unstable 0
```

enum vm_event_item global events (continued)

```
compact_fail 636
compact_success 471
compact_daemon_wake 972
compact_daemon_migrate_scanned 8420
compact_daemon_free_scanned 82677
htlb_buddy_alloc_success 0
htlb_buddy_alloc_fail 0
cma_alloc_success 0
cma_alloc_fail 0
unevictable_pgs_culled 13117
unevictable_pgs_scanned 768
unevictable_pgs_rescued 106
unevictable_pgs_mlocked 106
unevictable_pgs_munlocked 10
unevictable_pgs_cleared 0
unevictable_pgs_stranded 56
thp_fault_alloc 265372
thp_fault_fallback 15214
thp_fault_fallback_charge 0
thp_collapse_alloc 959
thp_collapse_alloc_failed 0
thp_file_alloc 0
thp_file_fallback 0
thp_file_fallback_charge 0
thp_file_mapped 32052
thp_split_page 17
thp_split_page_failed 0
thp_deferred_split_page 624
thp_underused_split_page 0
thp_split_pmd 827
thp_scan_exceed_none_pte 1161
balloon_inflate 0
balloon_deflate 0
balloon_migrate 0
swap_ra 67
swap_ra_hit 54
swpin_zero 75
swpout_zero 1292
ksm_swpin_copy 0
cow_ksm 0
zswpin 0
zswpout 0
zswpwb 0
direct_map_level2_splits 411
direct_map_level3_splits 24
direct_map_level2_collapses 0
direct_map_level3_collapses 0
nr_unstable 0
```

hardcoded zero after "NFS_Unstable:" node_stat_item removed in 2020

Events related to page allocation

- Physical (buddy) page allocations, for both userspace and kernel purposes
- Allocation success – **`pgalloc_normal`** (or **`_dma`**, **`_dma32`**, **`_movable`**, **`_device`**)
  - Could have been per-zone event instead? Yes, but… (probably historically)
- Also on success – **`numa_`** events (those being per-zone can substitute per-zone **`pgalloc`**)
  - **`numa_local`** from node of allocating cpu, otherwise **`numa_other`**
  - **`numa_hit`** when from preferred node (i.e. via mempolicy, implicitly local node)
    - **`numa_interleave`** when **`numa_hit`** and mempolicy is (weighted) interleave
  - **`numa_miss`** otherwise, and then also **`numa_foreign`** on the preferred node's zone
- Page freeing increments **`pgfree`** (no zone distinction)
- All of these count base pages (2^n per single allocation), except **`numa_interleave`**, oops!

Events related to page allocation

- Allocating means the free memory decreases, eventually below a watermark (min → low → high)
- Drops below the low watermark – kswapd woken up for the NUMA node
  - Kswapd initiates memory reclaim on its node – increments **`pageoutrun`** (global counter) and reclaims until free memory goes above the high watermark
- Kswapd tries to sleep for 100ms only
  - It's woken up during that short sleep – increments **`kswapd_low_wmark_hit_quickly`**
  - Not woken up but free memory went below the high watermark during the short sleep – increments **`kswapd_high_wmark_hit_quickly`** and continues reclaming
  - Not woken up, still above high watermark – goes sleep with no further timeout
- Free memory drops below min watermark – allocation must perform direct reclaim – increments **`allocstall_normal`** (**`_dma`** etc. depending the zone where the allocation was requested)
  - May be counted repeatedly if the attempt fails, or else allocates the freed memory faster
  - Too many concurrent direct reclaimers can make them sleep instead - **`pgscan_direct_throttle`**

Events related to page faults

- **`pgfault`** – page fault was handled (for whatever reason – mapping doesn't exist, or is read-only and userspace wants to write), unit is per fault regardless of folio size
- **`pgmajfault`** – disk I/O (read from swap or file) was necessary to finish the page fault
  - readahead can prevent it even for a first fault for given part of a file
- **`pswpin`** – reads from swap (unit is base page)
  - **`swap_ra`** – swap folio readahead (unit is per folio), **`swap_ra_hit`** – first fault (thus minor) on a readahead swap folio
- **`pgpgin`** – reads via the block layer, the unit is kilobytes, includes other sources than page faults
- **`pgreuse`** – fault on a write-protected page table entry that can simply make the entry writable using the already mapped page
  - either shared file mapping or the last of parent/child writing to a copy on write (CoW) page

Events related to memory reclaim

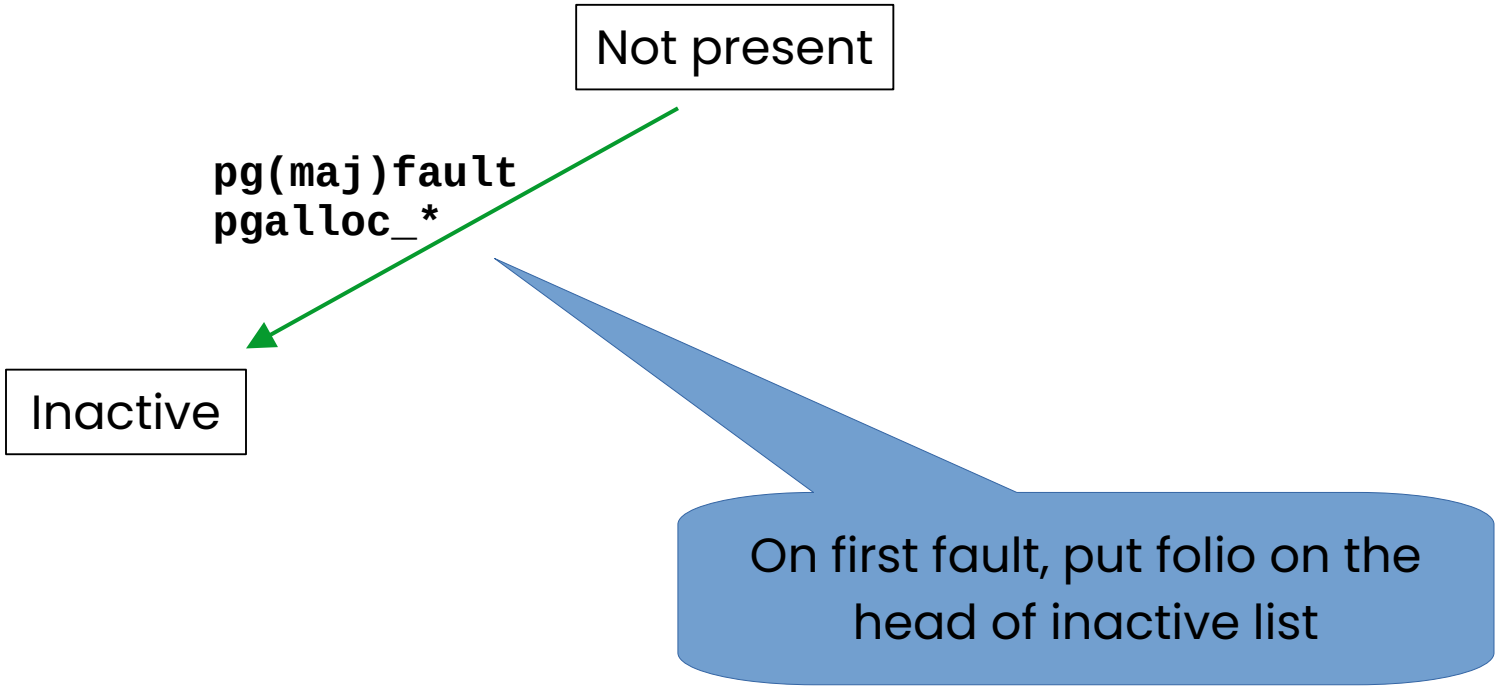
- Folios are organized in LRU lists, split into file/anon and active/inactive lists, more recently accessed folios closed to head of the list
  - The sizing of the active/inactive split is artificial, meant to hold most folios on the active list and focus reclaim on the smaller inactive list
    - Inactive ratio grows with square root of size of the LRU lists
    - With 1GB size, active:inactive is 3:1, with 10GB it's 10:1 etc…
- Reclaim calculates how many folios from each list to scan
  - To maintain the ratio, to account for `vm.swappiness`, recent reclaim "cost"…
  - Then scans the lists from the tail (where the coldest pages should be)
- The related events often distinguish **`_kswapd`**/**`_direct`**/**`_khugepaged`** (for deferred THP collapsing)/**`_proactive`** (/sys/…/nodeX/reclaim) and **`_anon`**/**`_file`**

Events related to memory reclaim

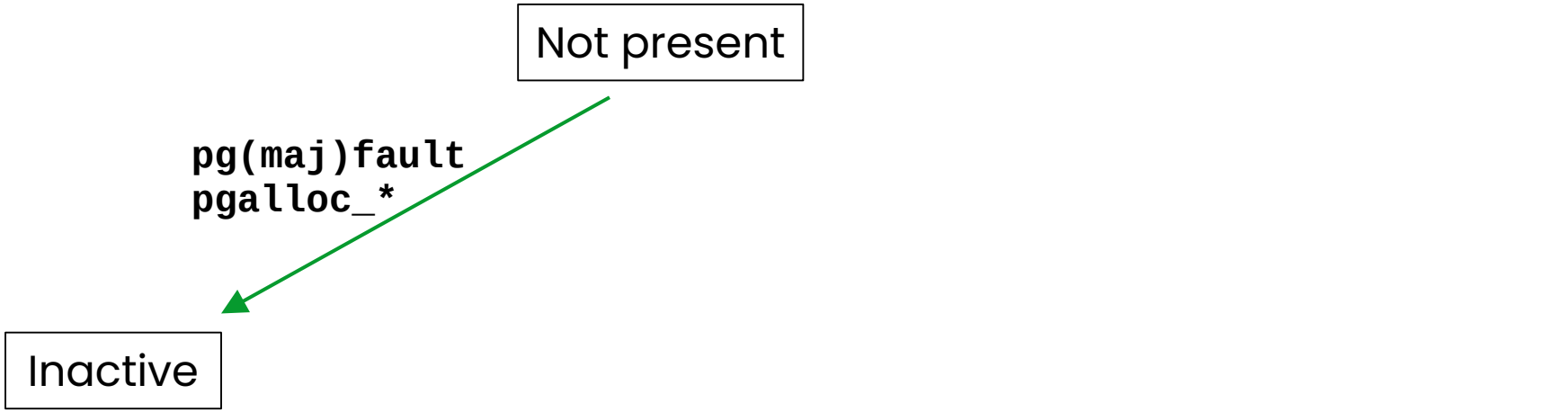
Not present

Events related to memory reclaim

promote

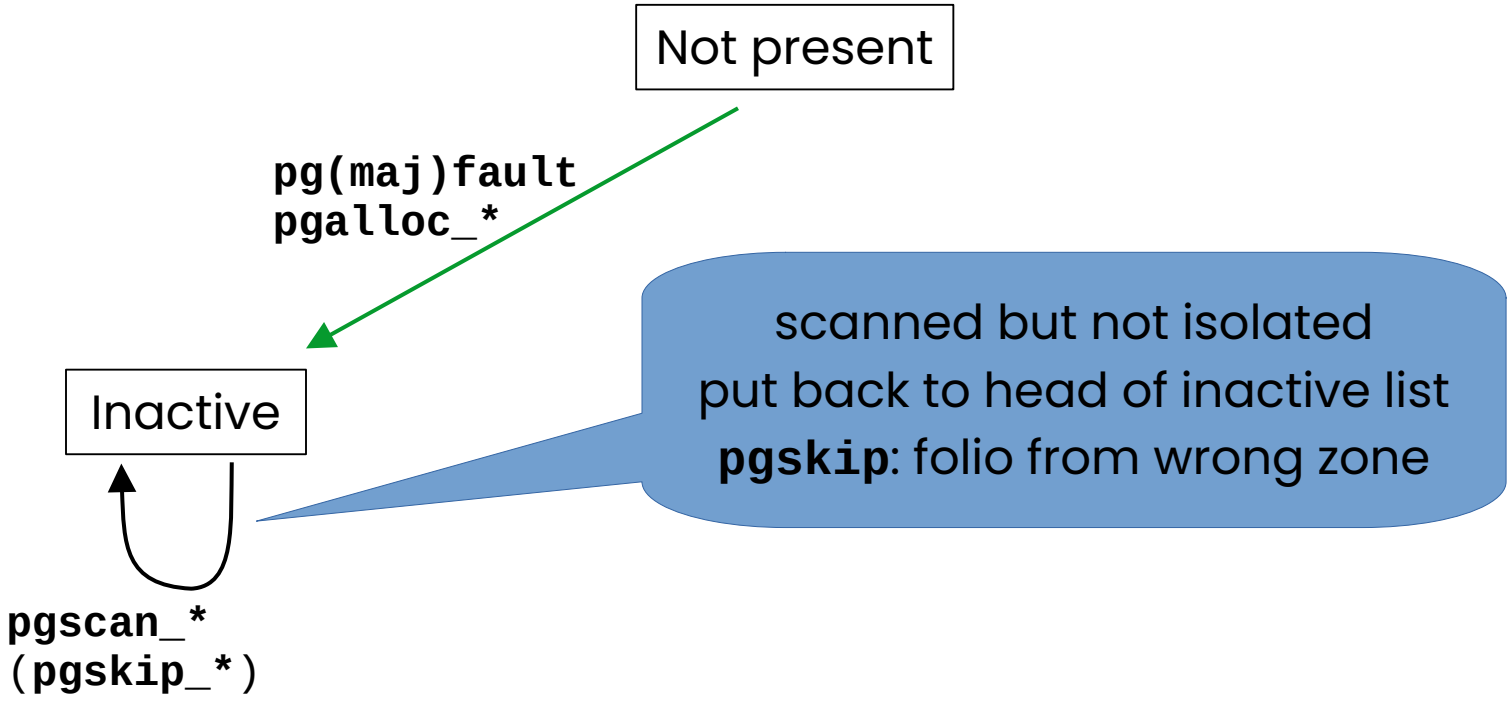


Events related to memory reclaim



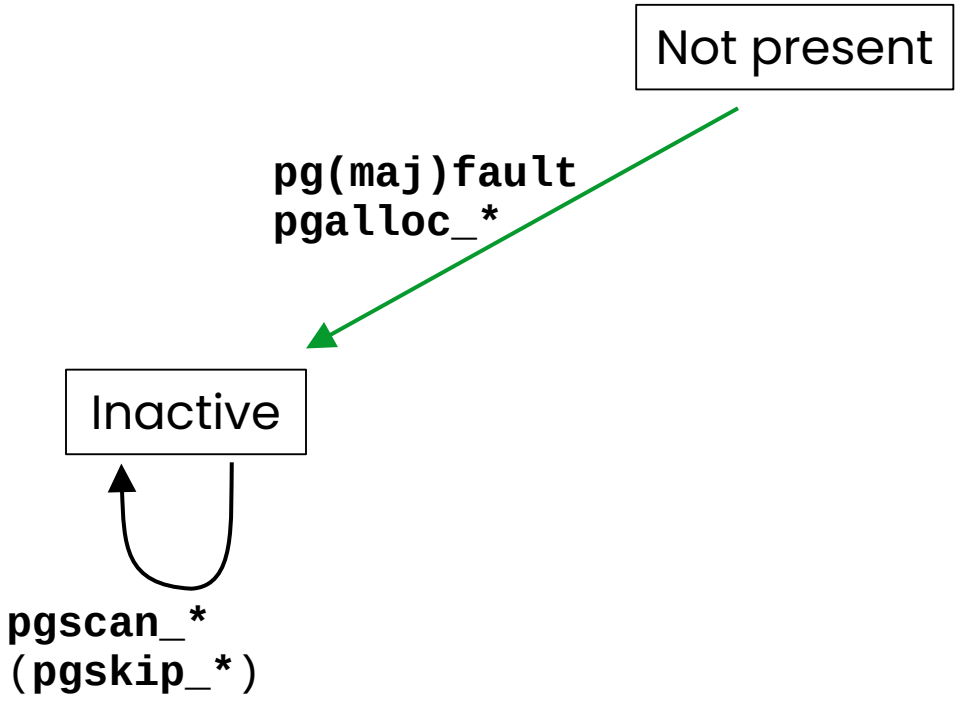
Events related to memory reclaim

promote

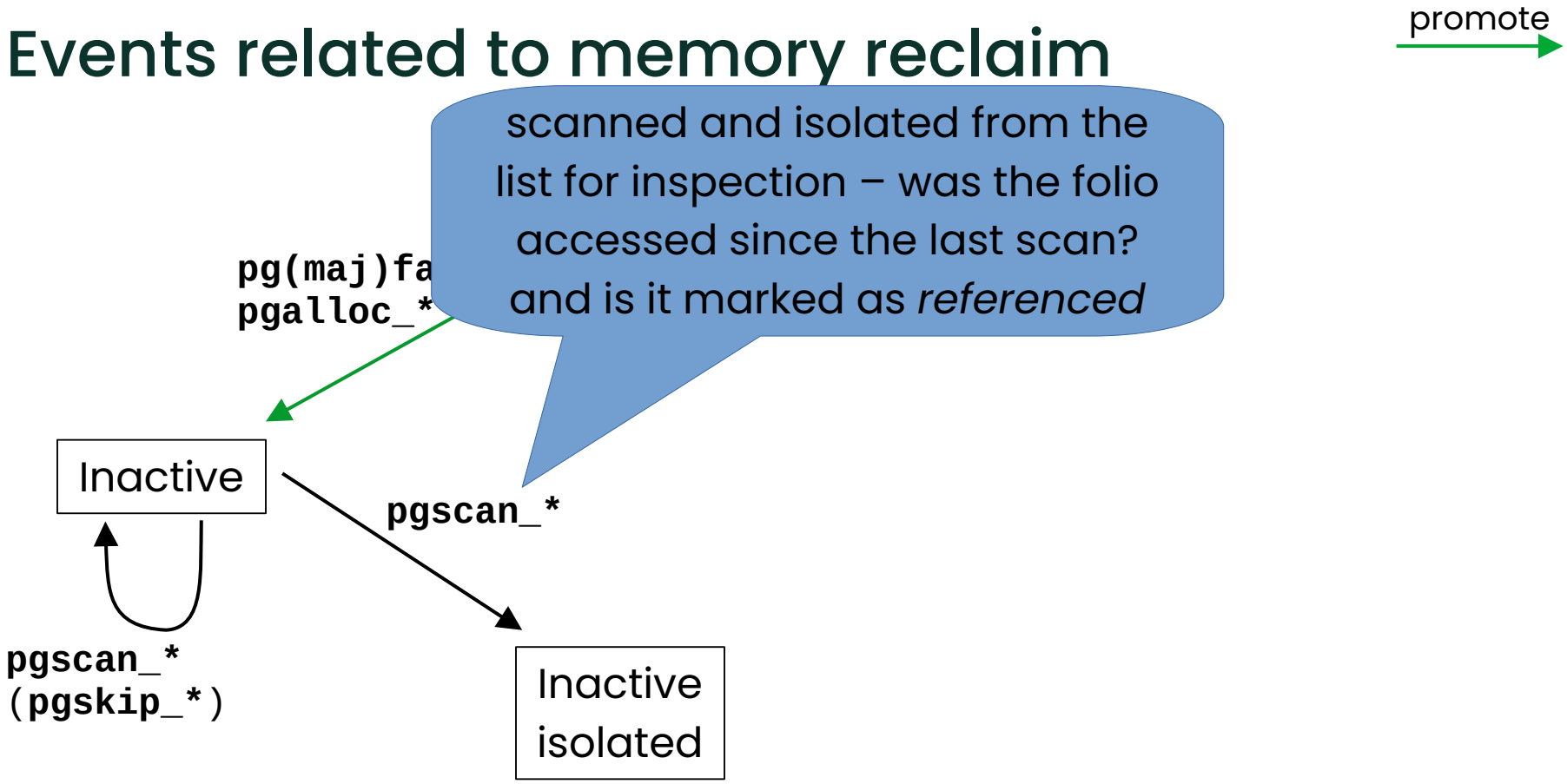


Events related to memory reclaim

promote

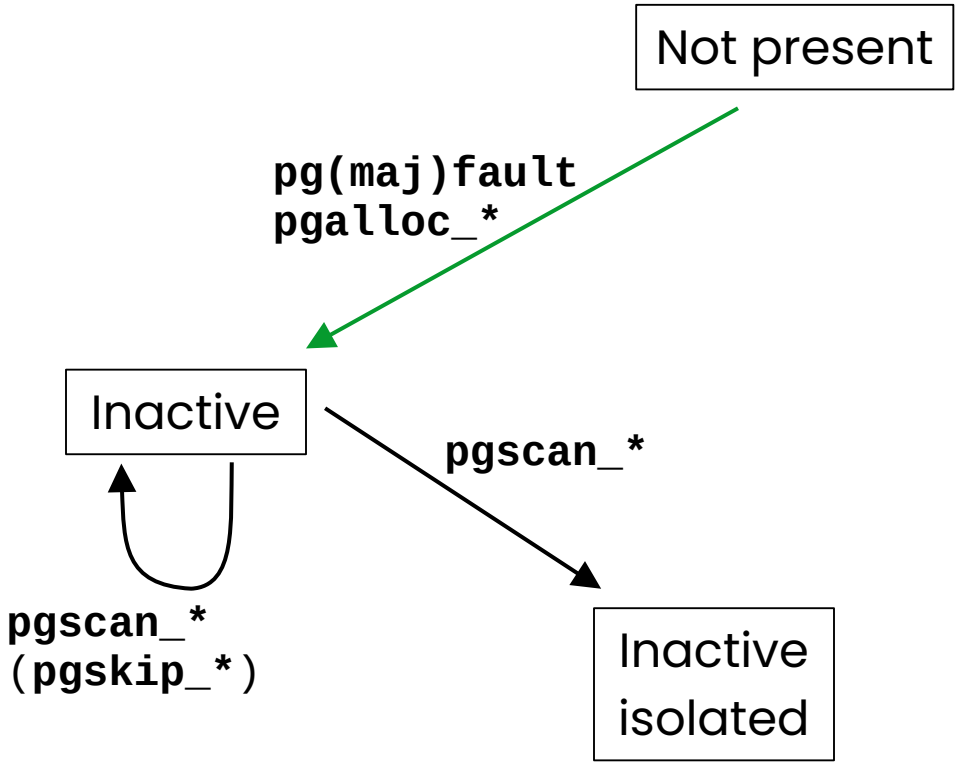


Events related to memory reclaim

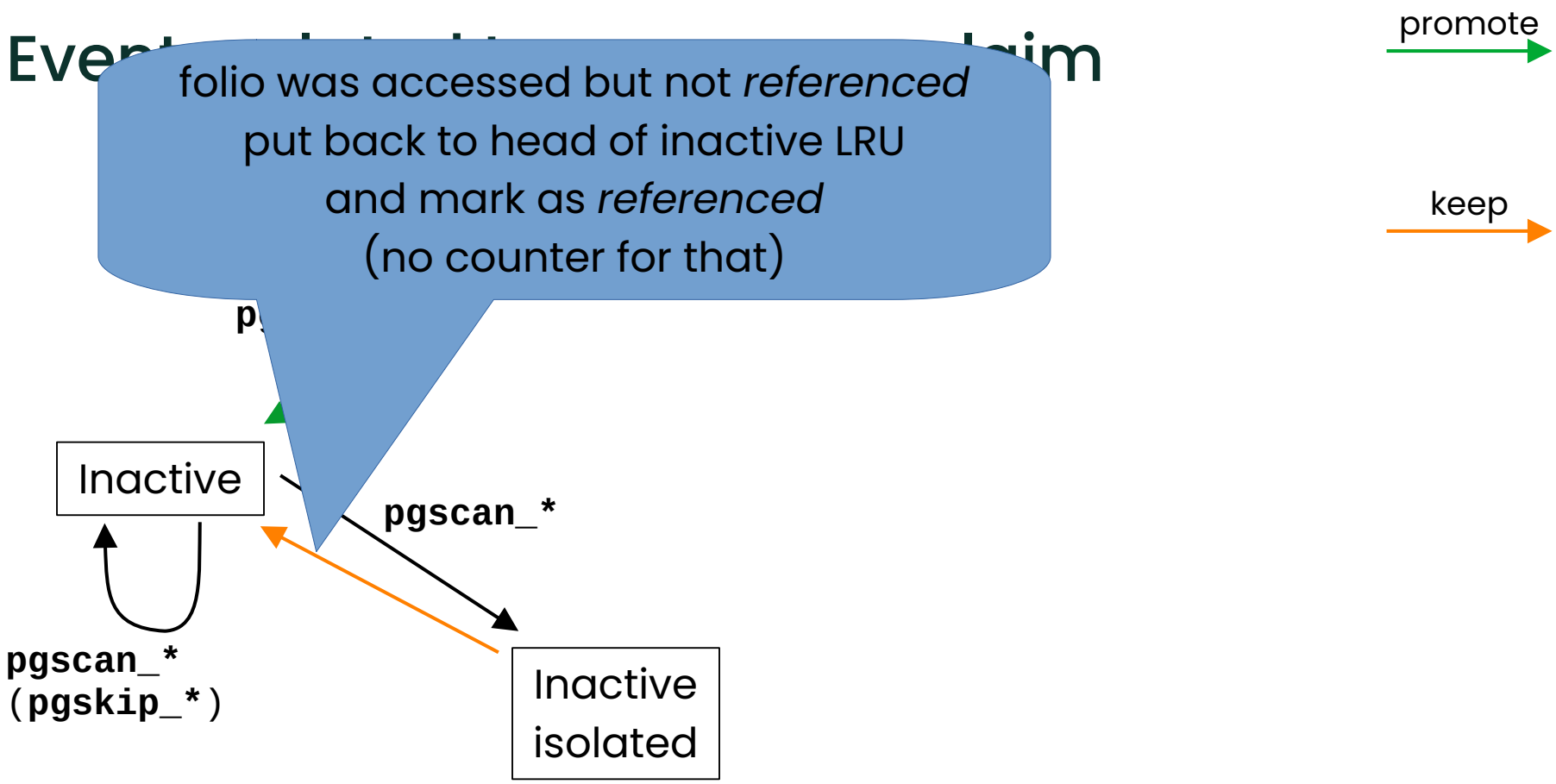


Events related to memory reclaim

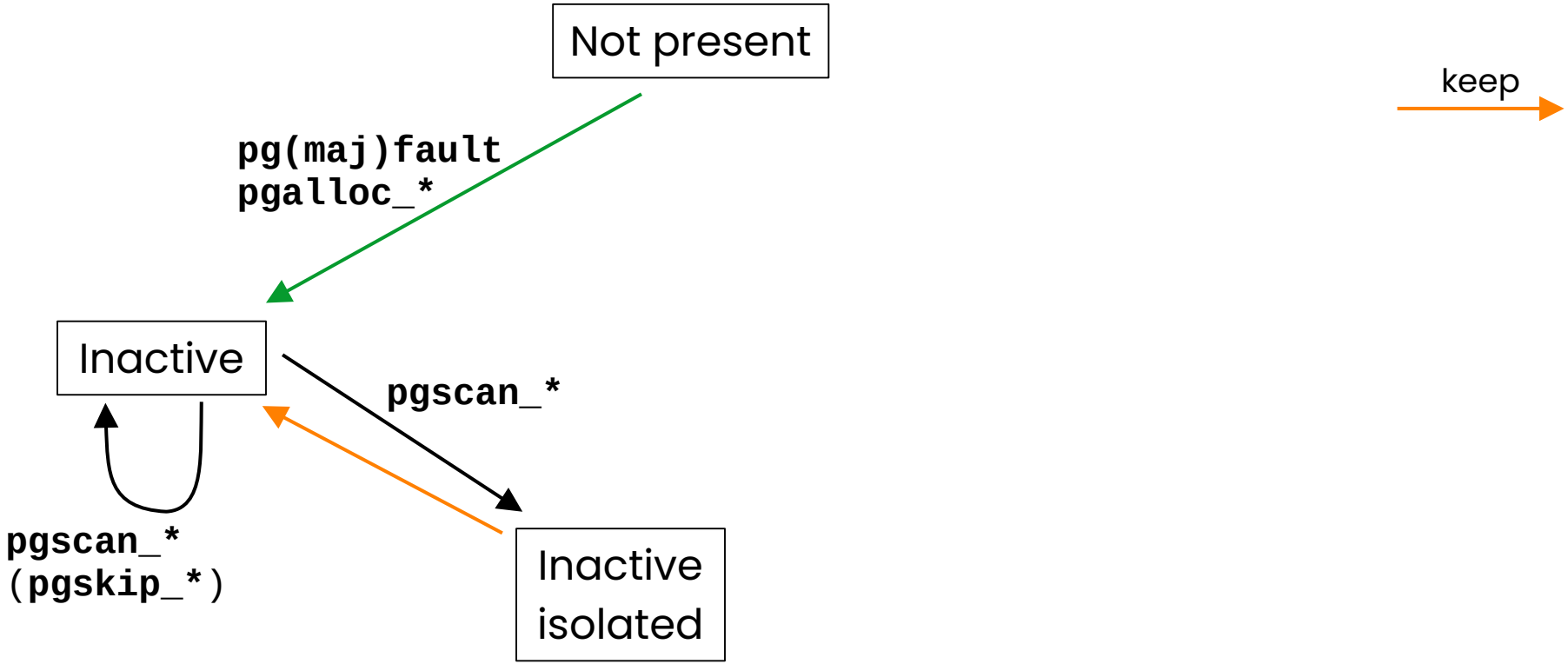
promote



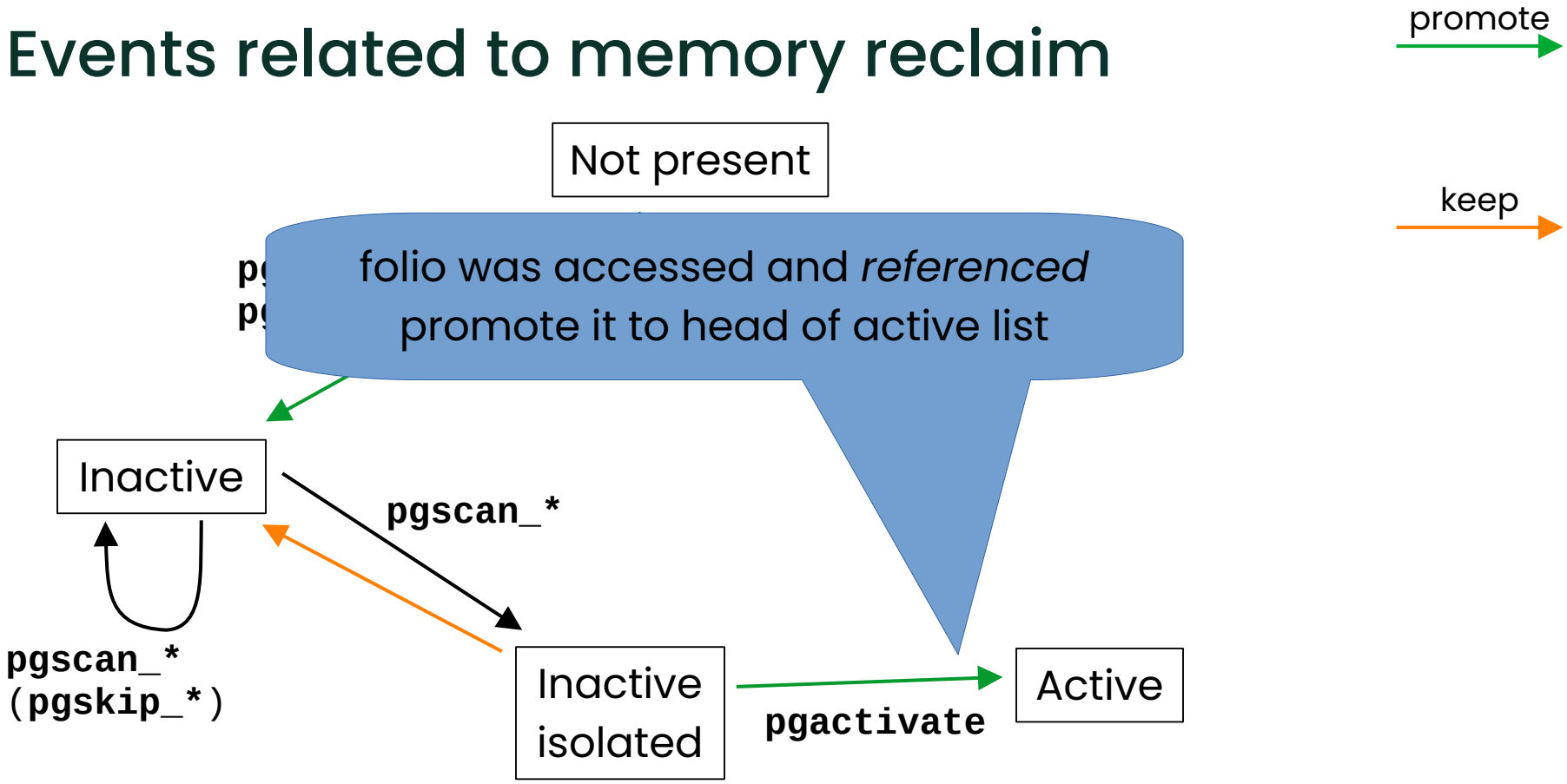
Even[illegible]aim


promote
keep
folio was accessed but not *referenced*
put back to head of inactive LRU
and mark as *referenced*
(no counter for that)
Inactive
pgscan_*
pgscan_*
(pgskip_*)
Inactive
isolated

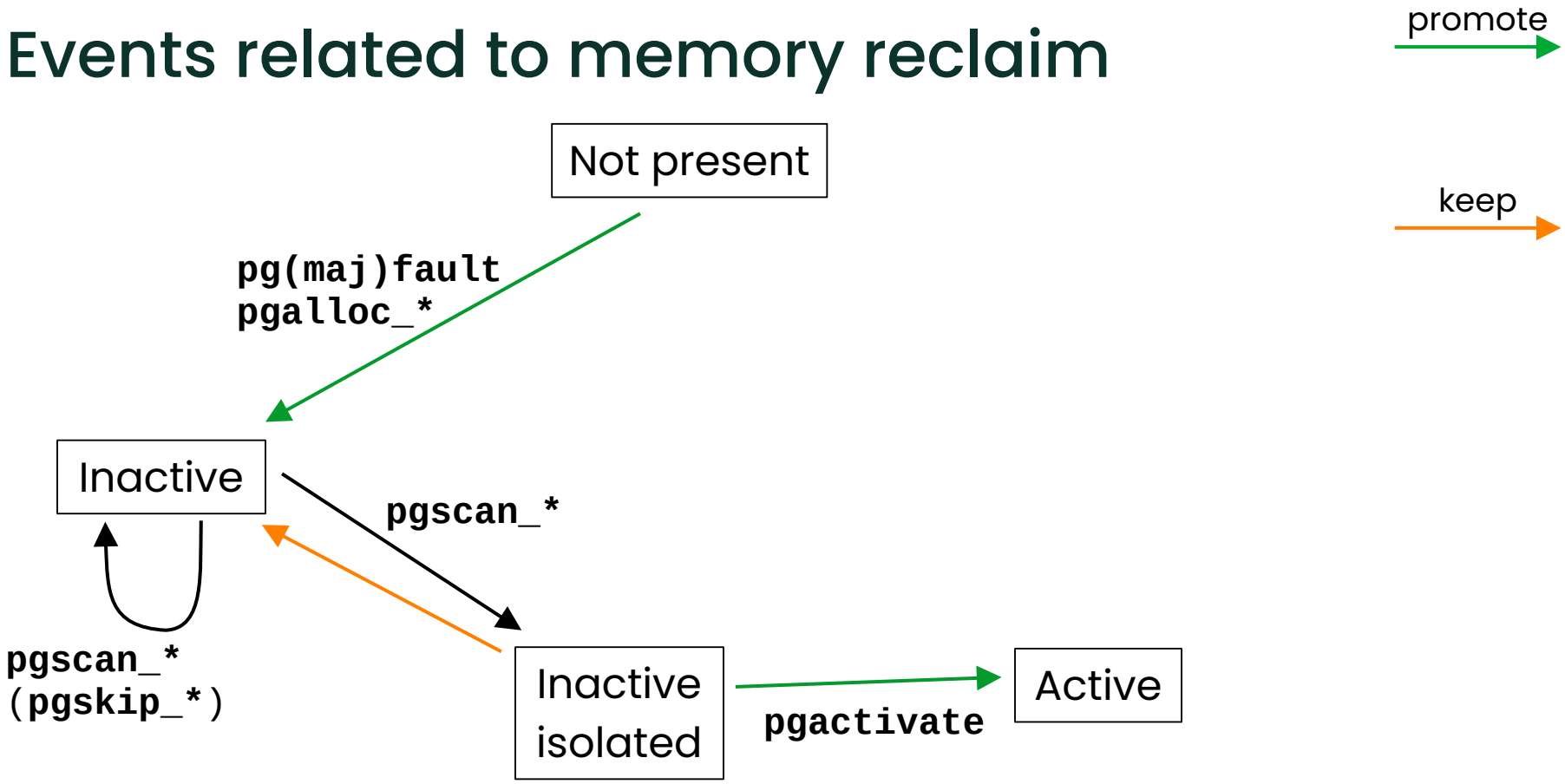
Events related to memory reclaim



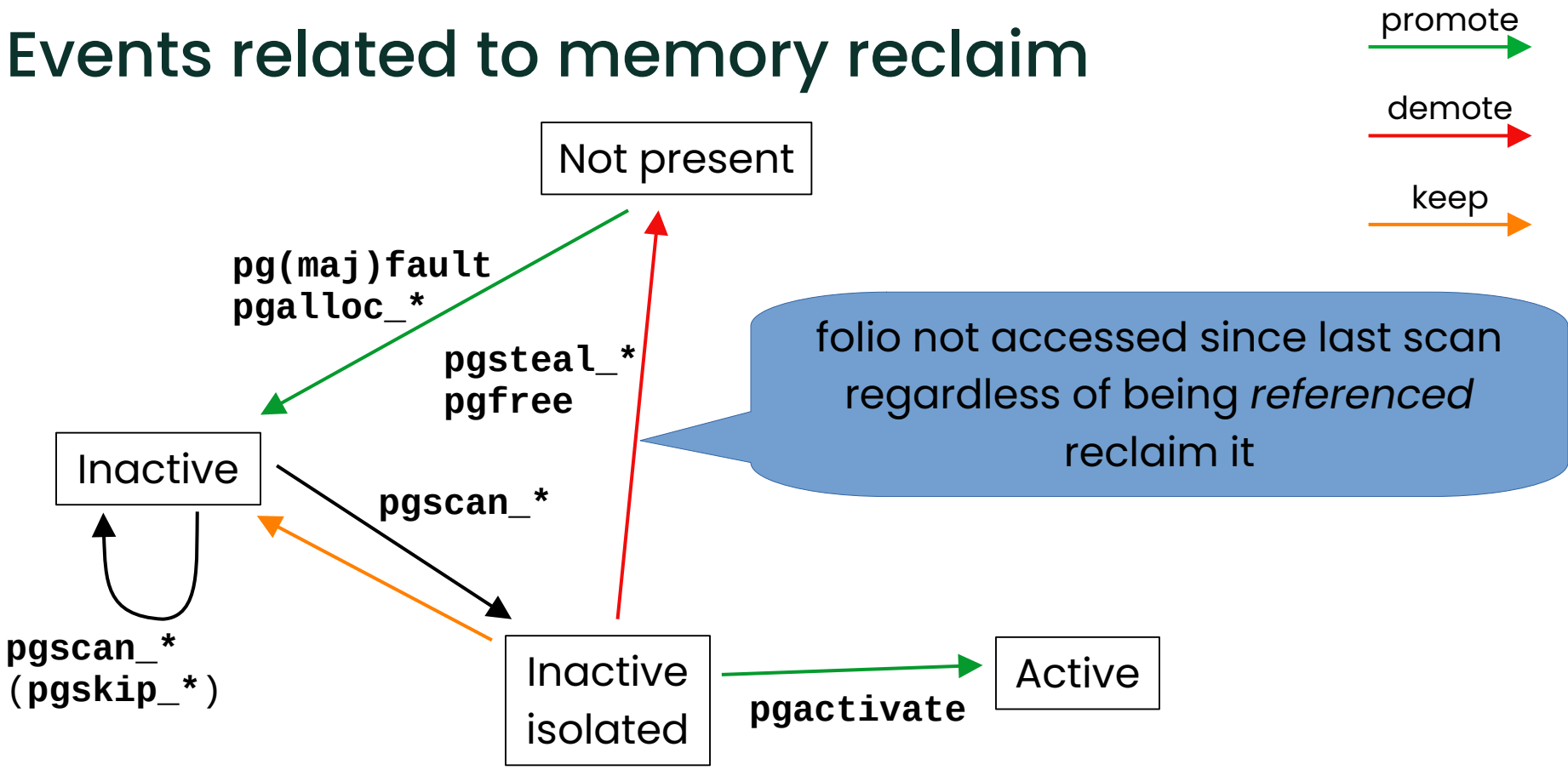
Events related to memory reclaim



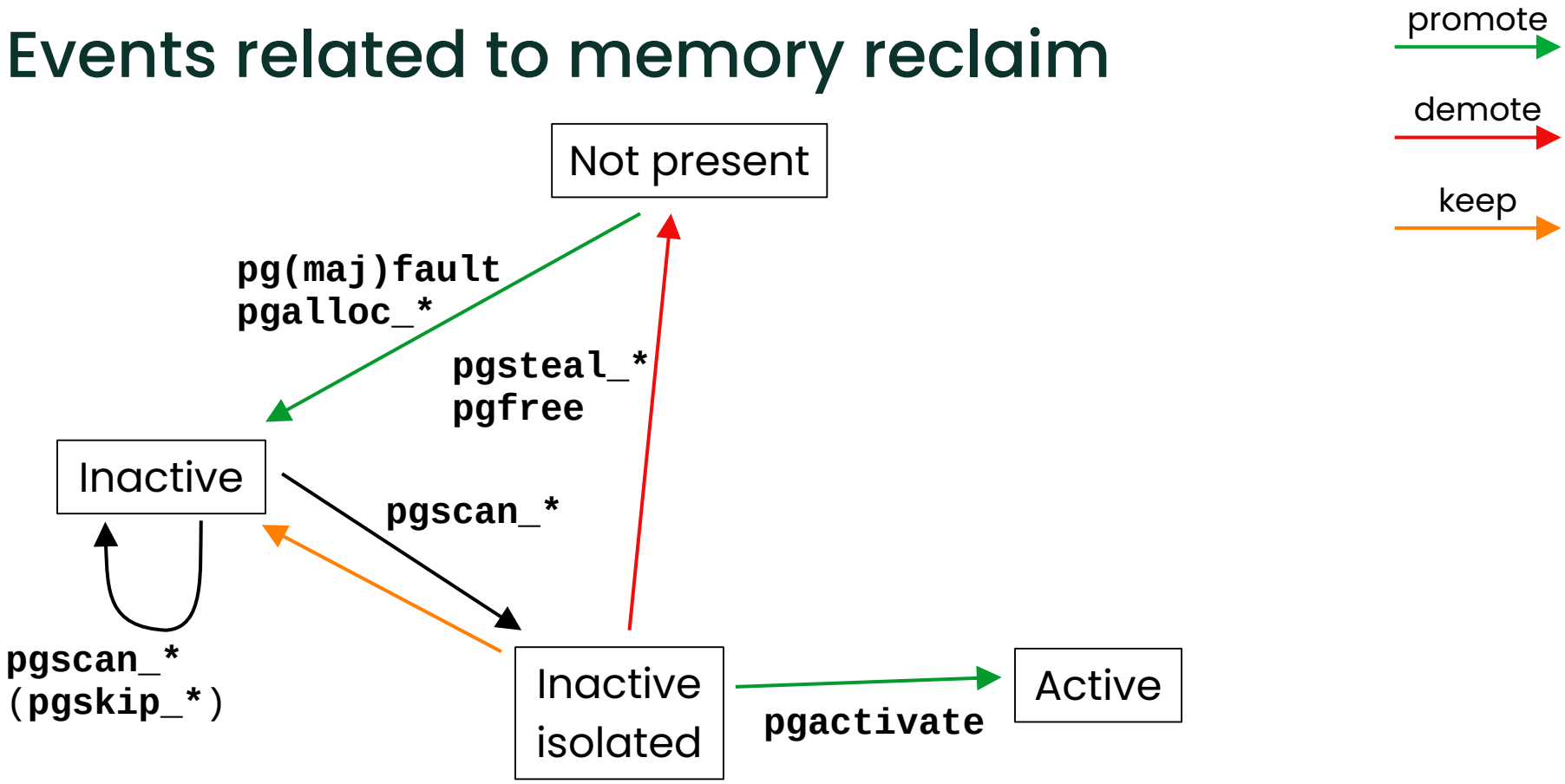
Events related to memory reclaim



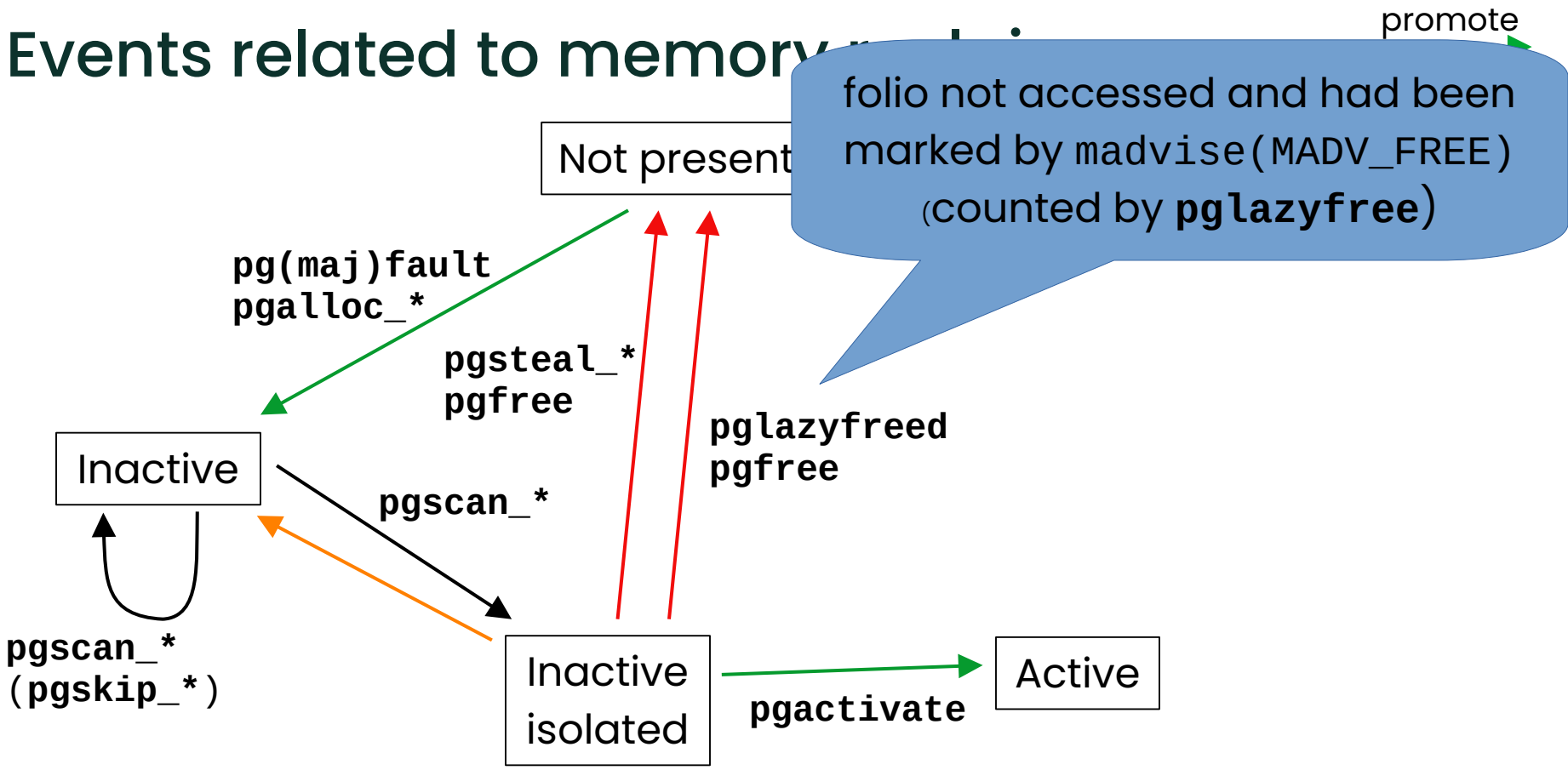
Events related to memory reclaim



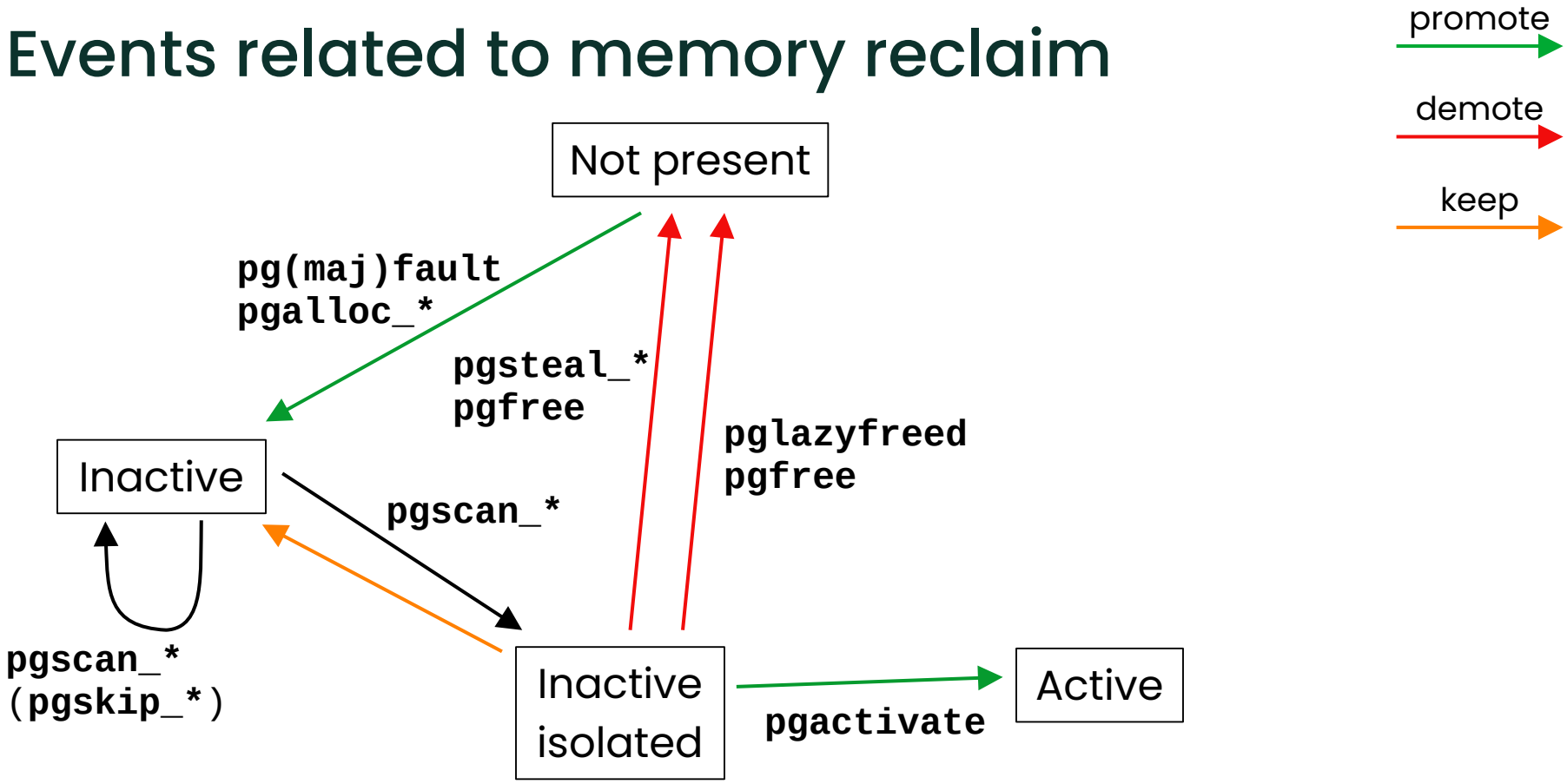
Events related to memory reclaim



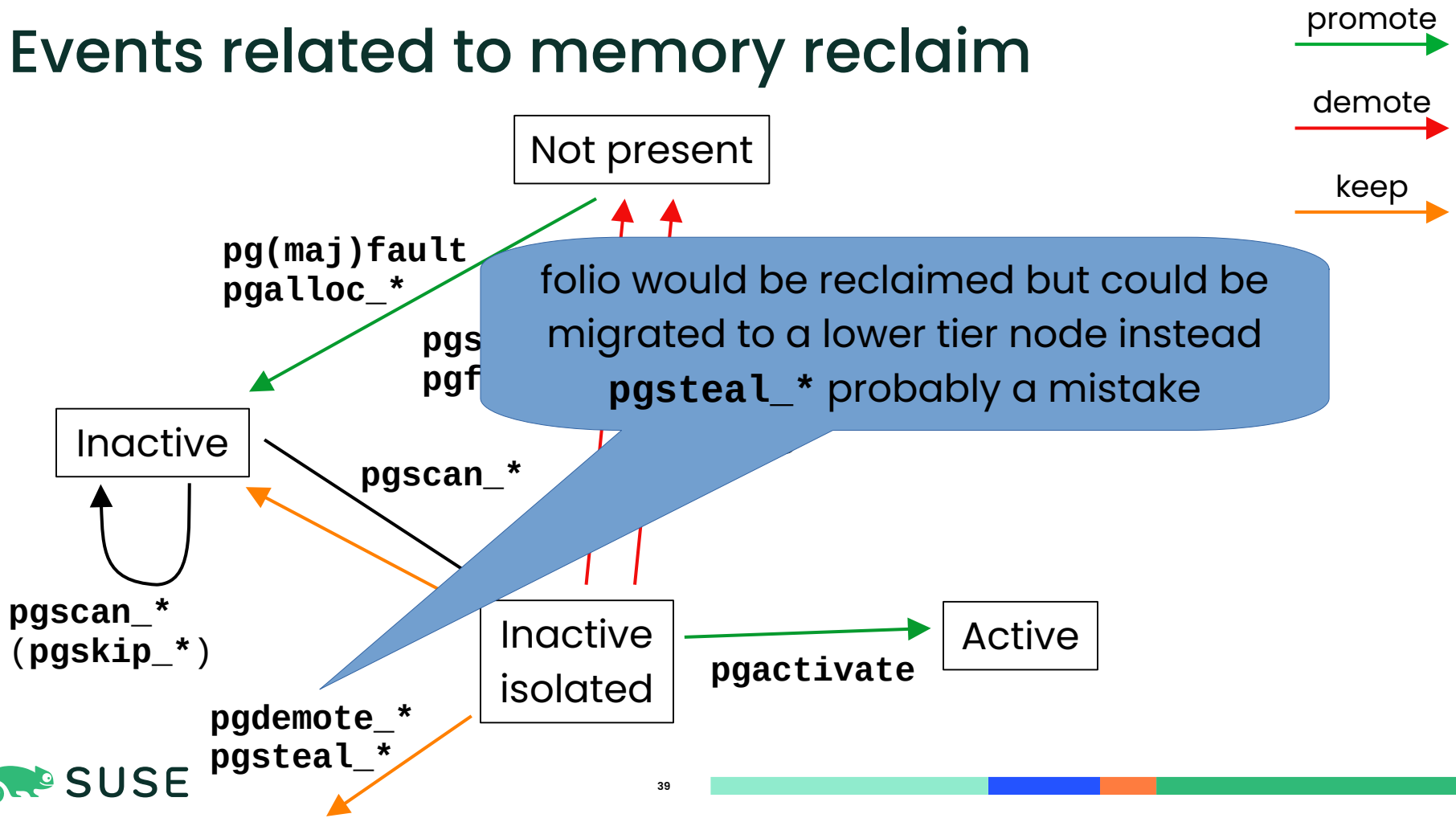
Events related to memory



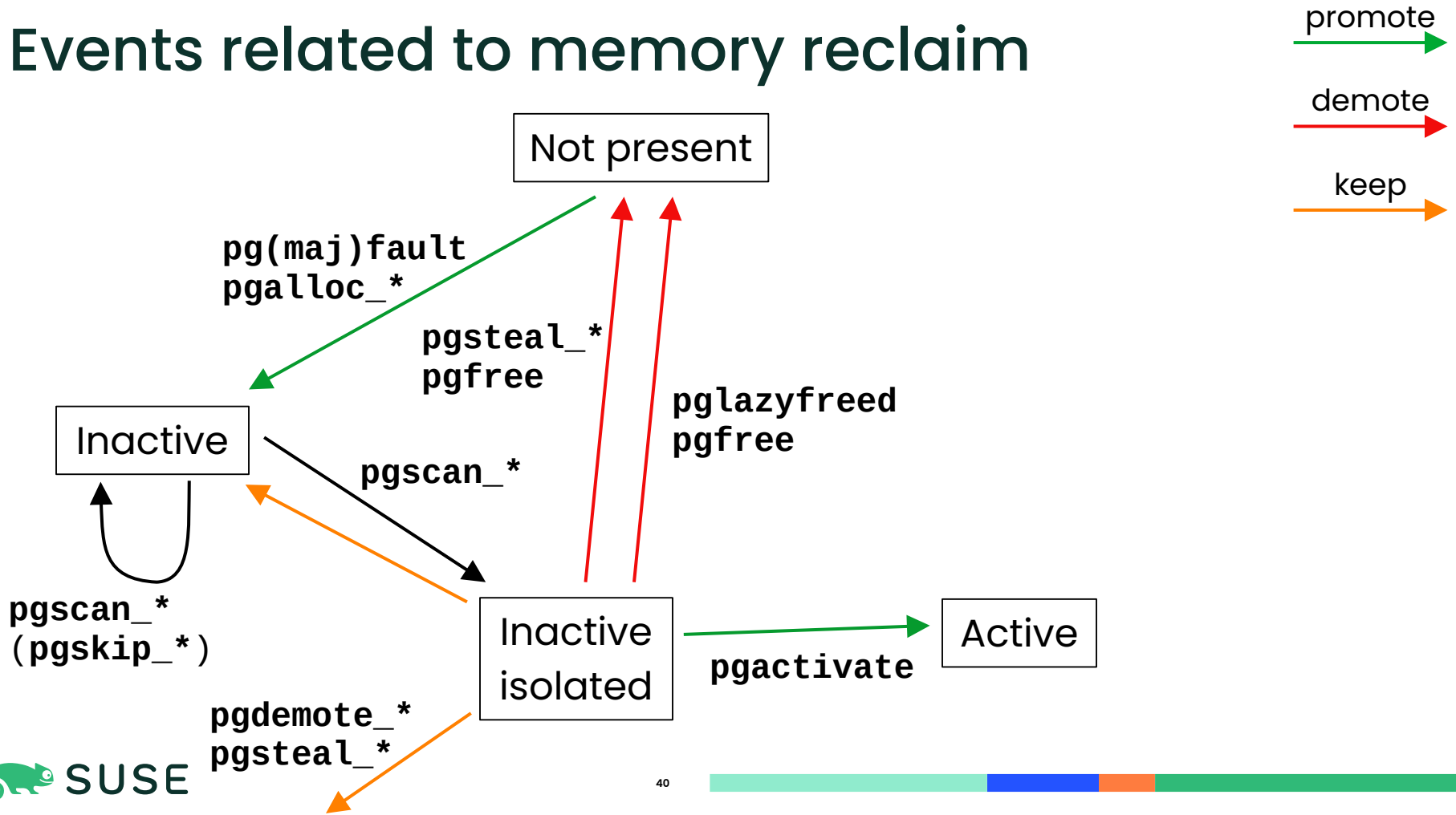
Events related to memory reclaim



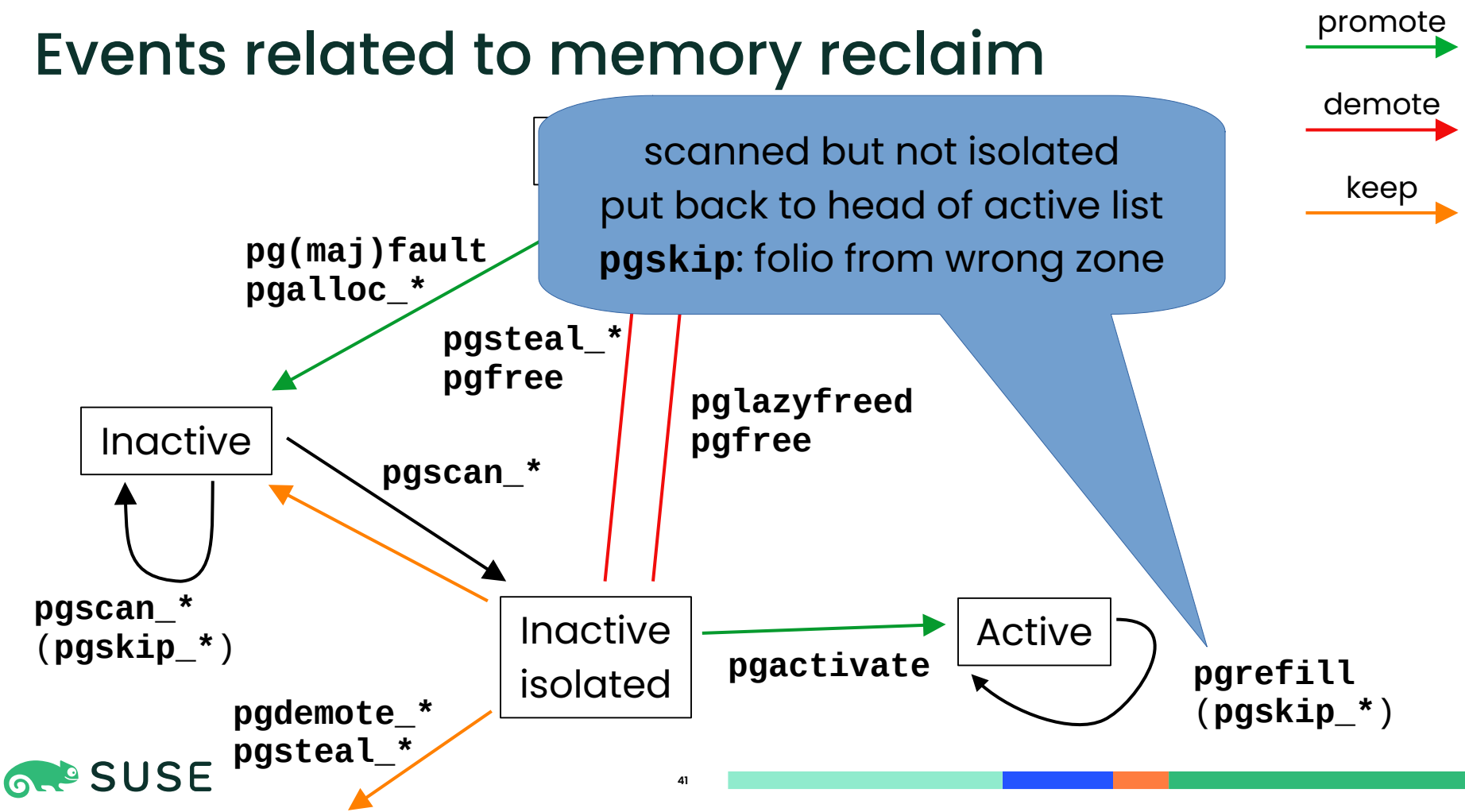
Events related to memory reclaim



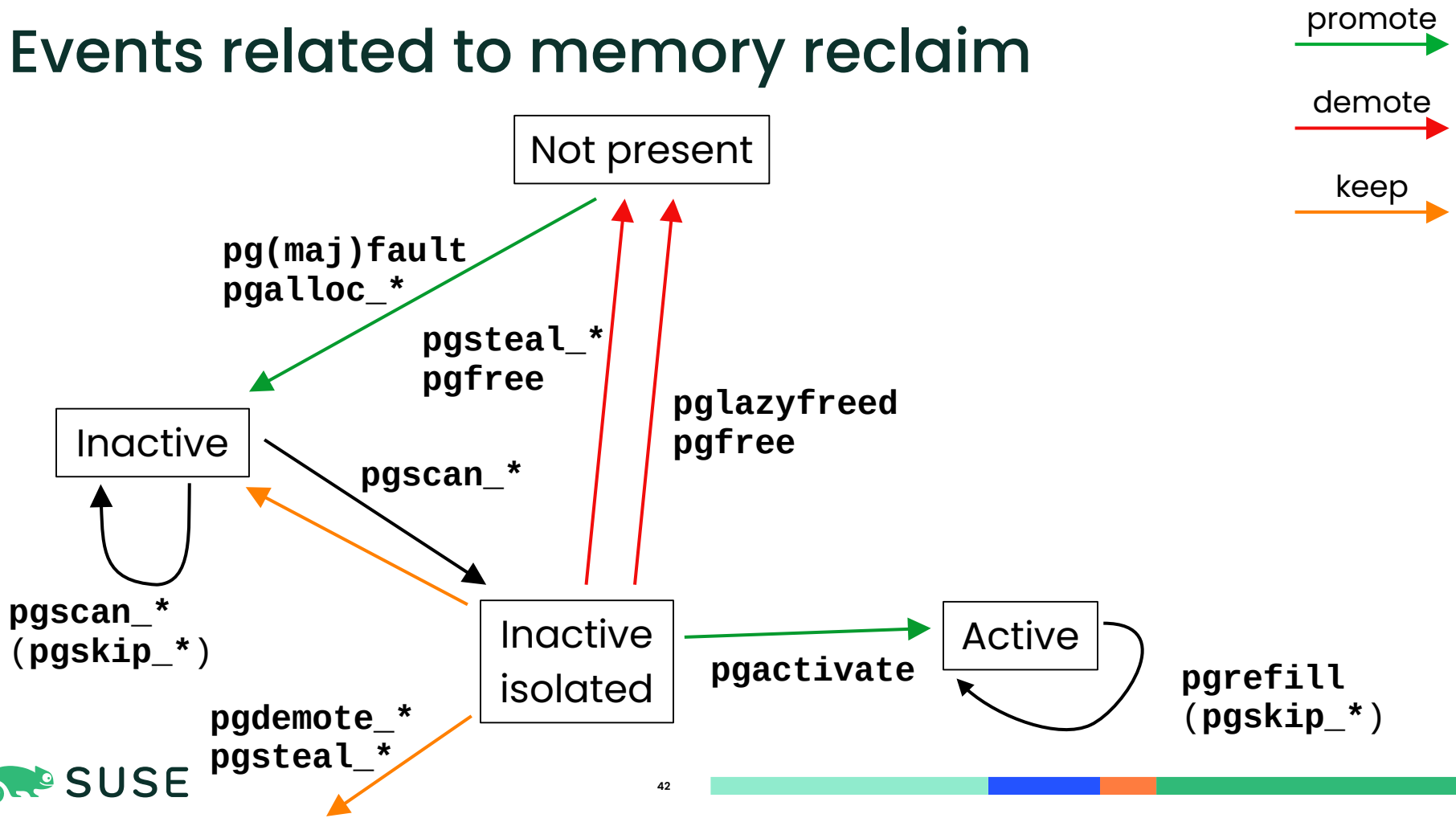
Events related to memory reclaim



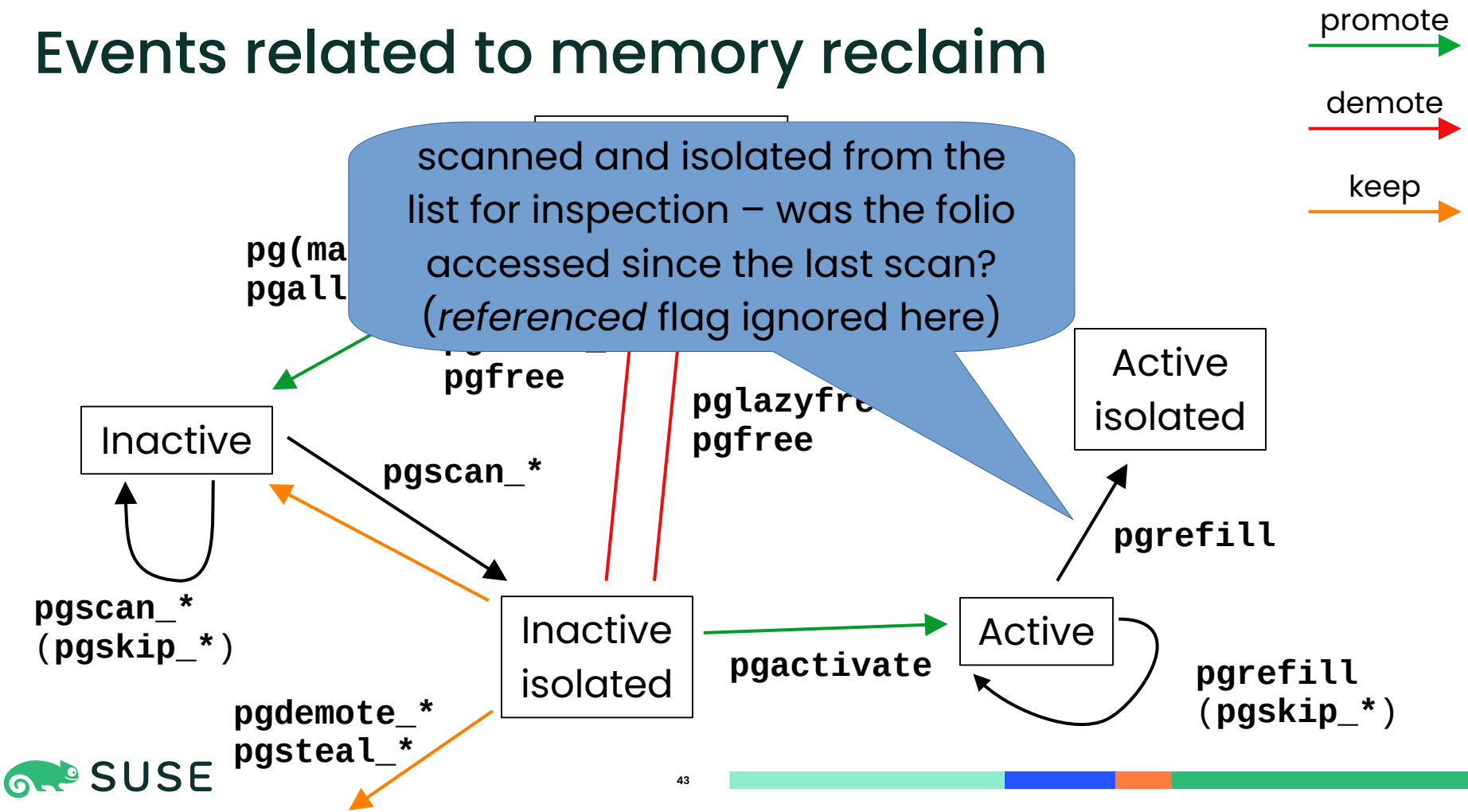
Events related to memory reclaim



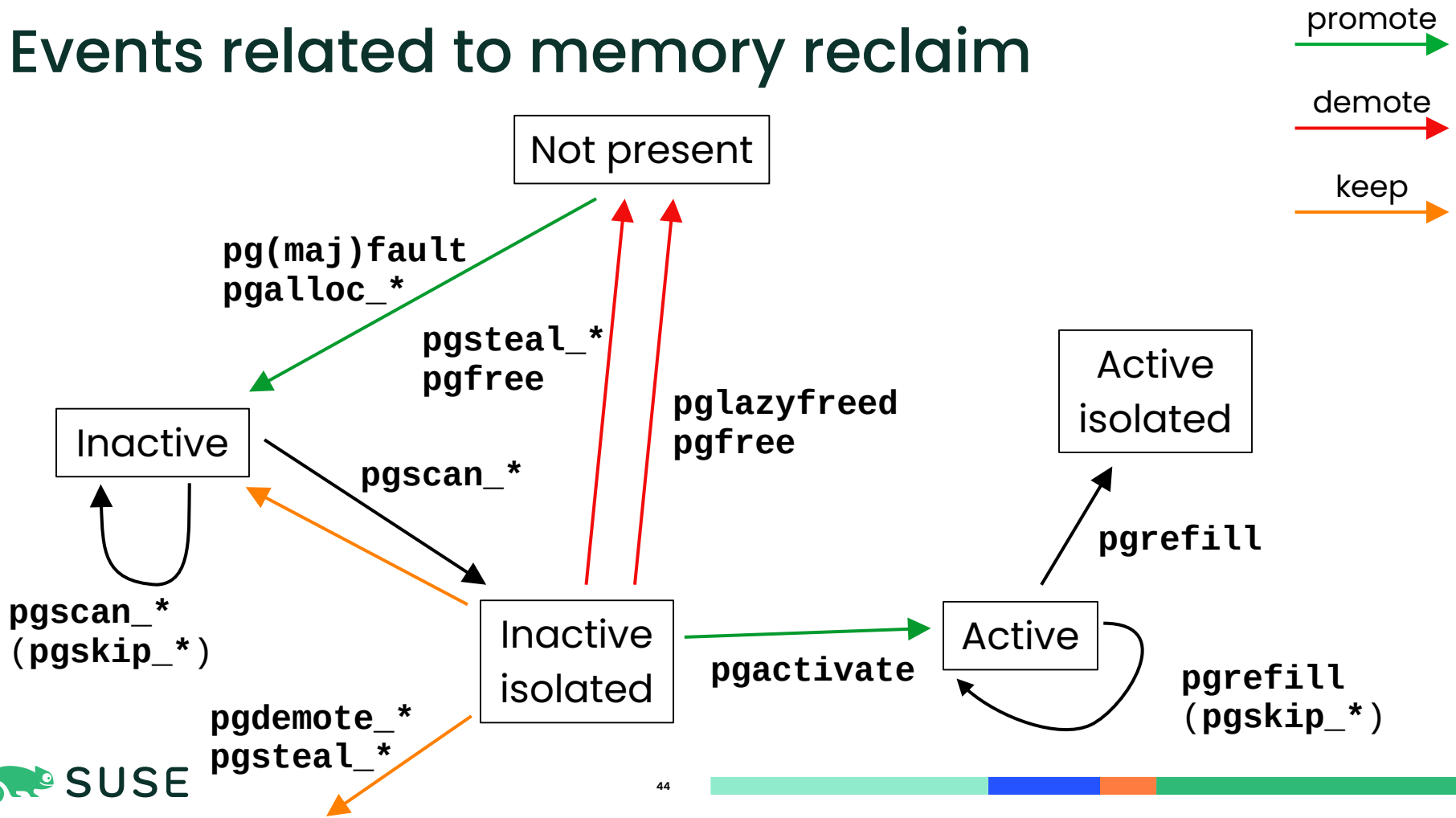
Events related to memory reclaim



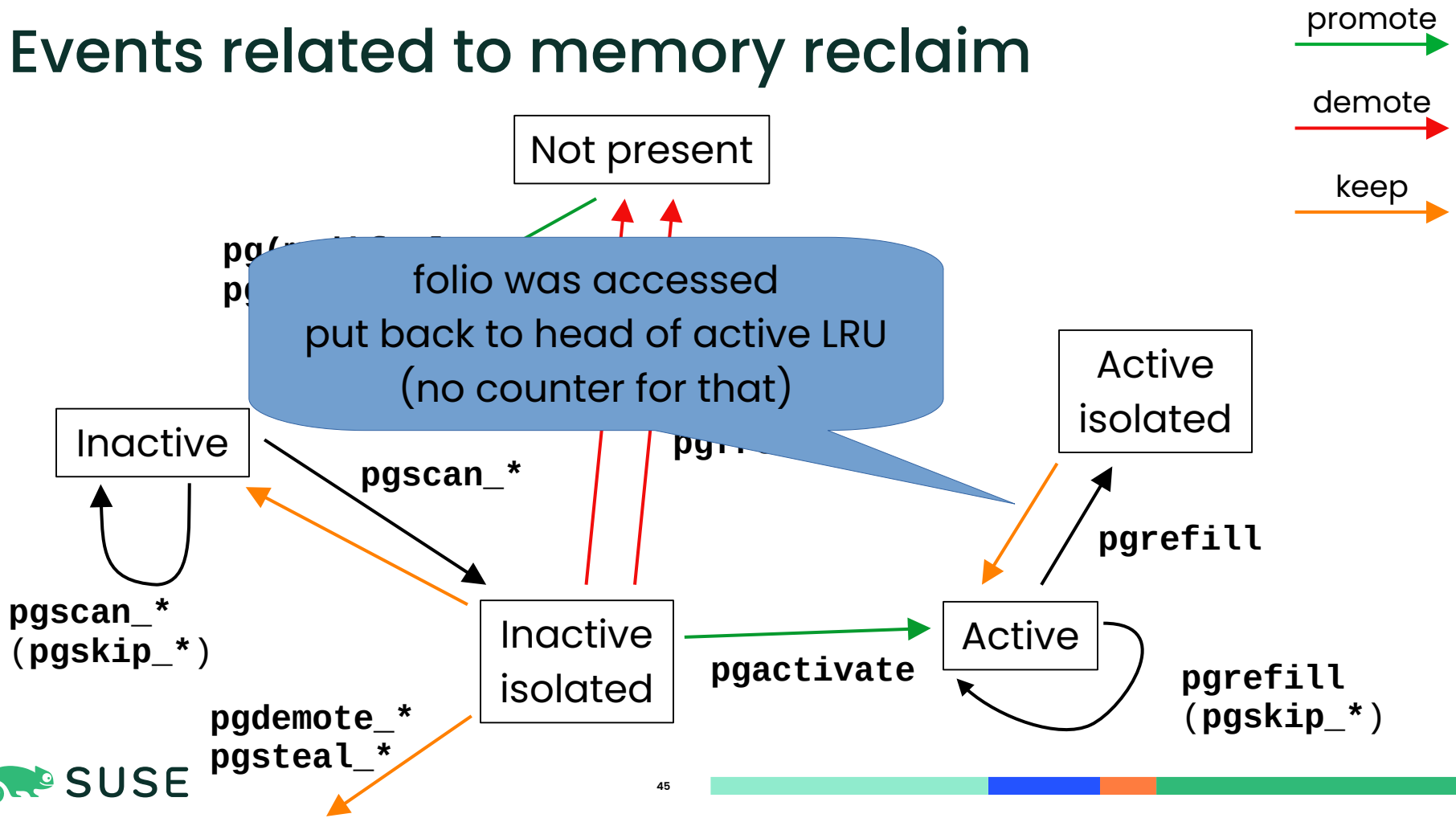
Events related to memory reclaim



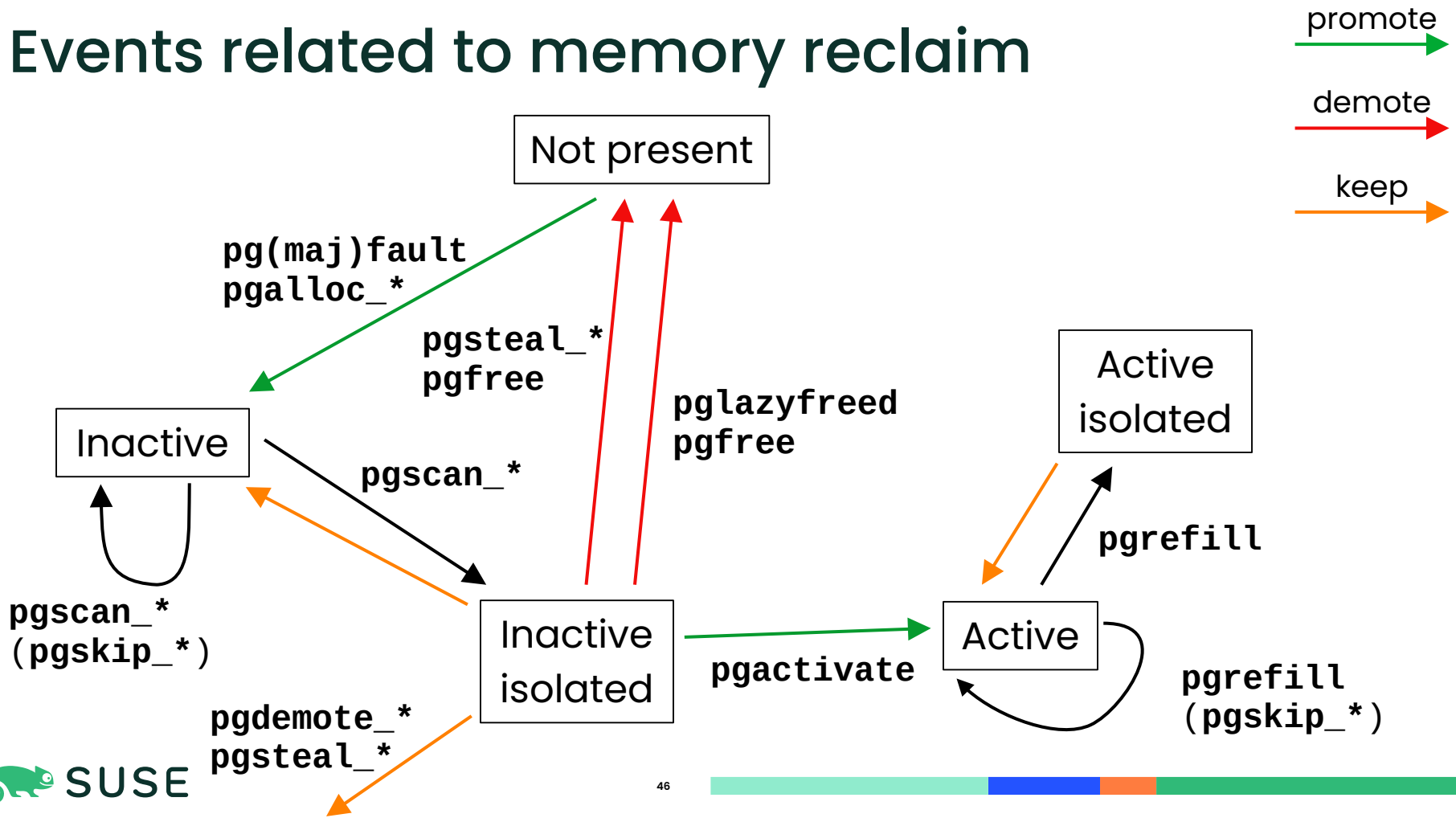
Events related to memory reclaim



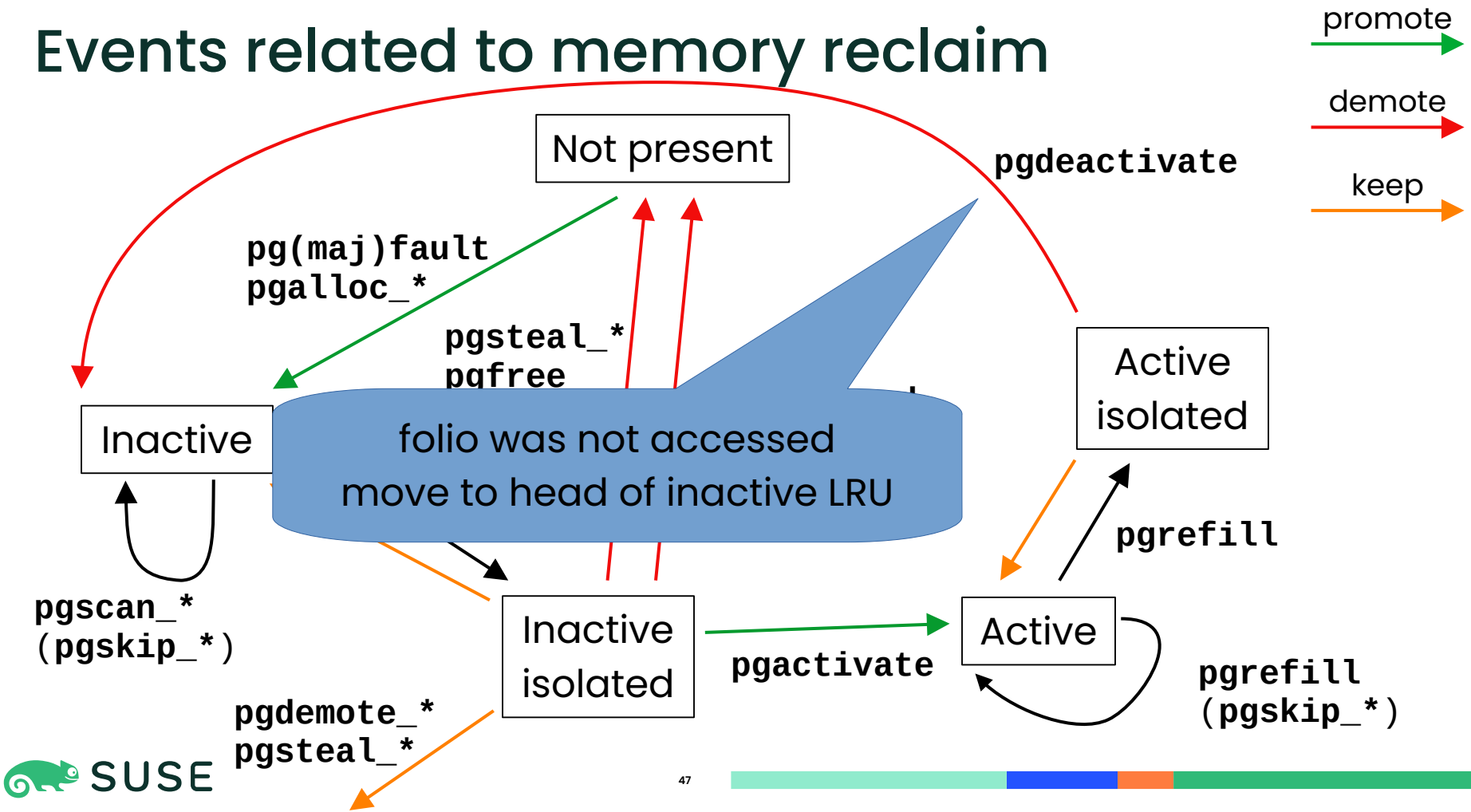
Events related to memory reclaim



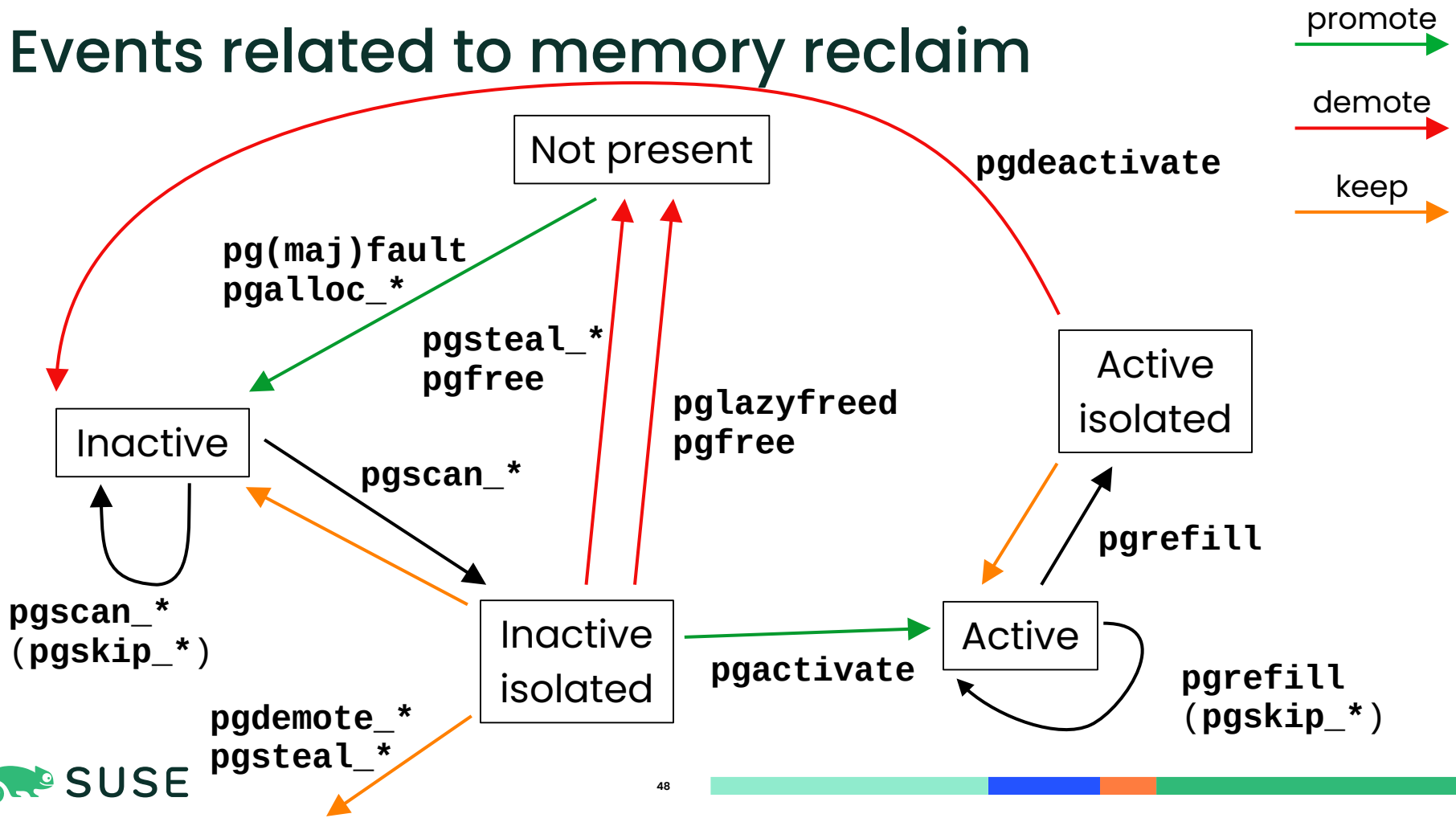
Events related to memory reclaim



Events related to memory reclaim



Events related to memory reclaim



Events related to reclaim/writeback

- anonymous and shmem folios go to swap and are written back immediately (if dirty)
  - **`pswpout`** – pages written to swap (unit is base pages) during reclaim
- dirty file folios are written back outside of reclaim (flusher kthreads) independently
  - reclaim will only mark dirty folios with a `reclaim` folio flag and put them on active list
    - so it doesn't encounter them again during another pass of inactive list
    - but this also increments the **`pgactivate`** counter! (bug?)
    - same after initiating with swap out - unless the write is synchronous
- **`pgrotated`** – incremented (unit is base pages) when folio writeback finishes (no longer dirty) and has the `reclaim` flag – it's moved to the tail of inactive list (immediately)
- **`pgpgout`** – all writes via the block layer, unit is kB, swap and writeback only part of it

Events related to workingset detection

- Motivation: workloads changing their working set (which memory they access) might be thrashing if folios can't be accessed often enough while on inactive list to have chance to be promoted to active list
  - Example: a system with only 4 or 8GB RAM running Firefox, now LibreOffice is started
  - Inactive list is intentionally small, the workload's new working set might be just larger
  - If a recently reclaimed folio is faulted in again, we don't know if it's new or thrashing
  - Meanwhile the folios on active list might be already idle, but we won't shrink it quickly enough
- Idea: remember when folios are reclaimed using "shadow nodes" that outlive them
  - Track certain events in reclaim to approximate "access distance" between two accesses to the same folio
  - Use this to estimate if the refaulted folio would have not been reclaimed in the first place, if the inactive list was larger on the expense of the active list, and also the other folio type's lists
    - I.e. for a file fault, would there have been reclaim+refault of this folio if the file inactive file list size would grow by the file active list size, and also both anon list sizes (if swapping is enabled)
    - If the answer is yes, put the page on active list instead of inactive list

Events related to workingset detection

- **`workingset_refault_anon`/`_file`** – a fault reads back a folio that was previously reclaimed (it has a shadow entry)
- **`workingset_activate_anon`/`_file`** – a refault happening soon enough after reclaim to consider the folio active – it will be put on active list immediately instead of inactive
- **`workingset_restore_anon`/`_file`** – an activation refault of a folio previously active
- **`workingset_nodes`** – current number of shadow nodes (not events)
- **`workingset_nodereclaim`** – shadow nodes reclaimed via slab shrinker

Conclusion

- /proc/vmstat events provide insight into what the mm subsystem is doing
- Helps for dealing with reports such as
  - “my system was slow and I think it was mm’s fault”
  - “kswapd/kcompactd takes too much cpu time”
  - “swap usage increased even though there was free memory”
- Often the absolute values are not interesting
  - Take a snapshot e.g. every 10 seconds (also with meminfo), post-process to calculate differences and then e.g. per-second rate of events and how it changes over time
  - Can’t offer exact debugging recipes here (experience helps a lot) but knowing what the counters report (the details matter) and how things work is a good start

Conclusion – broad example usage

- An example of things to look for in the “swap usage increased even though there was free memory” case
  - Do snapshots of nr_ counters confirm that memory went below watermarks?
  - Does it correlate with swap usage increase (only in meminfo) and **`pswpout`**
  - Is page cache becoming very low and anonymous memory usage high in the process?
    - swap out caused by genuine memory pressure due to anonymous memory
  - Not, but **`workingset_activate_file`** increases?
    - swap out caused by some anon memory being more idle than those files
    - none or negligible **`pswpin`** events confirms the anon memory was idle

Conclusion – need more details?

- vmstat events we didn't cover – compaction, migration, THP, `mlock(3)`, slab shrinking, hugetlb, CMA, KSM, zswap, direct mapping splits/collapses, ballooning
- What if things behave unexpectedly and we suspect real mm issue?
  - E.g. reclaim failures, unexplained high kcompactd cpu usage
- /proc/vmstat might not provide enough details for events and with coarse granularity
- The next level of getting details is observing trace events
  - e.g.: `trace_mm_migrate_pages(stats.nr_succeeded, stats.nr_failed_pages, stats.nr_thp_succeeded, stats.nr_thp_failed, stats.nr_thp_split, stats.nr_split, mode,reason);` - all with timestamp, process context…
  - Much more data to collect and postprocess (or process online using BPF hooks)
- Even more low-level – `drgn` debugger scripts (on live vmcore)



SUSE

Thank you

Questions?